



**EGIRE
ROBOTICS**

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FPV Final Exam Study Guide

420 Questions, Answers & Explanations

Read the concepts, then attempt the exam

Prepared by: EgireRobotics

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How to Use This Guide

Study first, then test yourself in the real exam

About the exam

- You get 50 questions chosen at random from the bank in this guide. Every attempt is different.
- The pass mark is 70%. You have up to 3 attempts, with a 24-hour wait after a failed attempt, and 90 minutes per attempt.
- The exam runs in fullscreen and you cannot go back to an earlier question, so answer carefully before you move on.
- The order of the answer options is shuffled in the exam, so learn the answers, not the letters.

How to study

1. Take one topic at a time, in the order given below.
2. Read each question and try to answer it before looking at the green answer.
3. Read the "Why" explanation under every question, including the ones you got right. It explains the concept and gives a real example.
4. When a topic feels comfortable, move on, and come back to the questions you missed.
5. Attempt the exam only when you can explain the reasons, not just recall the answers.

Topics in this guide

1. Brushless Motor Physics — 50 questions
2. Electronic Speed Controller (ESC) Mastery — 50 questions
3. Flight Controller (FC) Architecture — 50 questions
4. Propeller Aerodynamics — 50 questions
5. Video Transmitter and Camera Systems — 50 questions
6. FPV Assembly and Soldering — 50 questions
7. FPV Professional Pilot Certification — 120 questions

Reading the questions

The correct option is shown in green with a tick. The gold-edged "Why" box under each question explains the concept and gives a practical example. After the exam, the questions you miss are shown to you again with the same explanations.

1. Brushless Motor Physics

How a brushless motor works: phases, magnets, KV, torque, Back EMF and what makes motors run hot or fail.

50 questions

MOTOR-001

What would happen if one motor phase becomes disconnected?

- A. Motor operates normally **B. Motor loses synchronization ✓** C. Motor becomes more efficient D. Battery charges

Why: A brushless motor needs all three phases energised in sequence to create a rotating magnetic field. With one phase disconnected the field can no longer rotate smoothly, so the rotor pulses, stutters and loses synchronisation with the ESC. Example: after a crash a motor wire tears off its solder pad — on take-off that motor twitches instead of spinning and the quad flips.

MOTOR-002

What component wears out in a brushed motor?

- A. Magnets B. Shaft **C. Brushes ✓** D. Windings

Why: Brushed motors pass current through carbon brushes that rub against a commutator, so the brushes physically wear away. Example: a brushed tiny-whoop motor loses power after a few months and needs replacing, while a brushless motor with no rubbing contacts lasts for years.

MOTOR-003

Which battery type is commonly used in FPV drones?

- A. NiCd B. Lead Acid **C. LiPo ✓** D. AA Battery

Why: LiPo (lithium polymer) batteries pack a lot of energy into very little weight and can deliver the high current bursts an FPV motor needs. NiCd, lead-acid and AA cells are too heavy or cannot supply that current. Example: a 4S 1300mAh LiPo weighs about 150g yet can deliver over 100A in bursts.

MOTOR-004

Higher KV motors generally provide:

- A. Higher RPM ✓** B. Higher Torque C. Lower Speed D. More Weight

Why: KV means RPM per volt, so a higher-KV motor spins faster on the same voltage — but with less torque. Example: a 2700KV motor on a 4S pack (about 16V) turns roughly 43,000 RPM unloaded, ideal for small fast racing props.

MOTOR-005

What determines the sequence of magnetic field rotation?

- A. ESC Phase Switching Order ✓** B. Battery Brand C. Propeller Color D. Frame Material

Why: The order in which the ESC energises the three motor phases (its phase switching order) decides how the magnetic field rotates around the stator, and so which way and how fast the rotor follows. Example: swapping two motor wires changes that order and the motor spins the opposite direction.



MOTOR-006

Why are brushless motors ideal for FPV drones?

- A. High Efficiency B. High Power-to-Weight Ratio C. Fast Response Time **D. All of the Above ✓**

Why: Brushless motors are efficient (little energy lost as heat), have an excellent power-to-weight ratio and respond very quickly to throttle changes — all three matter on an FPV quad, so the answer is "all of the above". Example: a 30g 2207 motor can produce around 1.5kg of thrust and change speed within milliseconds.

MOTOR-007

Why does motor temperature increase?

- A. Copper Losses B. Iron Losses C. Bearing Friction **D. All of the Above ✓**

Why: Motor heat comes from three sources: copper losses (current heating the winding wire), iron losses (magnetic losses in the stator core) and bearing friction. Example: after a long full-throttle freestyle pack the motors feel hot — if they are too hot to touch, check for a bad bearing or an over-propped build.

MOTOR-008

Copper windings are located on the:

- A. Rotor B. Shaft **C. Stator ✓** D. Bearings

Why: The copper windings are wound around the stator, the stationary part of the motor. The rotor (bell) carries the magnets instead. Example: open a motor and the fixed inner core wrapped in copper wire is the stator; the outer bell that spins holds the magnets.

MOTOR-009

Why can't a brushless motor run directly from a battery?

- A. Wrong Connector **B. Needs 3-Phase Switching ✓** C. Too Heavy D. No Magnets

Why: A battery supplies steady DC, but a brushless motor has no brushes to switch the current between coils. Something must switch the DC into a three-phase sequence — that is the ESC's job. Example: touching a battery directly to the motor wires would just lock the rotor in one position or short-circuit.

MOTOR-010

What does a motor shaft do?

- A. Stores Energy **B. Transfers Rotation ✓** C. Measures Voltage D. Creates GPS Signals

Why: The shaft carries the rotor's rotation out of the motor so the propeller can turn. Example: the prop nut tightens onto the motor shaft; after a hard crash a bent shaft shows up as heavy vibration on the video.

MOTOR-011

Who generates the 3-phase power?

- A. Receiver **B. ESC ✓** C. Flight Controller D. Battery

Why: The ESC (electronic speed controller) turns the battery's DC into the three-phase drive for the motor by rapidly switching MOSFETs. The receiver, flight controller and battery cannot do this. Example: a 4-in-1 ESC board contains four of these three-phase drivers, one per motor.

MOTOR-012

A brushless motor requires:

- A. AC-like 3-phase power ✓ B. Direct DC connection C. USB power D. RF power

Why: A brushless motor needs three-phase, AC-like power: three drive signals 120° apart that the ESC creates from the battery's DC. Plain DC or USB/RF power cannot produce a rotating field. Example: the ESC raises motor RPM simply by switching through the three phases faster.

MOTOR-013

Heavy-lift drones generally use:

- A. High KV Small Motors B. Low KV Large Motors ✓ C. Brushed Motors D. Tiny Whoop Motors

Why: Heavy-lift drones carry big propellers and heavy payloads, so they need high torque at low RPM — which means large, low-KV motors. Example: a 15-inch agricultural drone typically uses motors around 100–400KV on a high-voltage battery.

MOTOR-014

What does KV mean in drone motors?

- A. Kilovolt B. RPM per Volt ✓ C. Current Rating D. Motor Efficiency

Why: KV stands for RPM per volt — it is a speed constant, not kilovolts or a current rating. Example: a 2000KV motor on 10V should ideally reach about 20,000 RPM with no load.

MOTOR-015

What is the main disadvantage of brushed motors?

- A. No Magnets B. Brush Wear ✓ C. No Rotation D. Too Efficient

Why: A brushed motor's brushes rub on the commutator, wearing out and creating friction and sparks, which limits its life. Example: cheap brushed micro-drone motors slowly lose power and must be replaced.

MOTOR-016

How many wires are connected to a brushless FPV motor?

- A. 2 B. 3 ✓ C. 4 D. 5

Why: A typical FPV brushless motor has three wires, one for each electrical phase (A, B and C). Example: you solder three motor wires to the ESC pads for each motor; there are no extra sensor wires on standard sensorless motors.

MOTOR-017

What is the primary purpose of a drone motor?

- A. Store Power B. Generate Video C. Convert Electrical Energy into Mechanical Rotation ✓ D. Process GPS Data

Why: A drone motor converts electrical energy from the battery into mechanical rotation that spins the propeller. It does not store power or handle video or GPS. Example: 150W of electrical power in might give about 120W of mechanical power at the shaft, the rest becoming heat.



MOTOR-018

Why is Back EMF important?

- A. Determines Rotor Position ✓ B. Improves Video C. Charges Battery D. Measures GPS

Why: Back EMF (the voltage produced by the spinning magnets) lets a sensorless ESC work out where the rotor is, so it knows exactly when to switch the next phase. Example: at very low RPM back EMF is weak, which is why motors can stutter or desync at the start of a throttle punch.

MOTOR-019

What is Back EMF?

- A. Battery Voltage B. Generated Voltage from Rotating Motor ✓ C. ESC Voltage D. GPS Signal

Why: Back EMF is the voltage generated in the windings by the rotating magnets, and it opposes the voltage you apply. Example: spin a motor by hand while its wires are connected to a multimeter and you will see a small voltage — the motor is acting as a generator.

MOTOR-020

Permanent magnets are attached to:

- A. Stator B. Rotor Bell ✓ C. ESC D. Propeller

Why: The permanent magnets are glued to the inside of the rotor bell, the outer part that spins. The stator holds the copper coils, not magnets. Example: the strong magnets inside the bell will pull a small screwdriver towards them.

MOTOR-021

What is motor cogging?

- A. Smooth Rotation B. Magnetic Resistance During Rotation ✓ C. Battery Failure D. ESC Desync

Why: Cogging is the notchy magnetic resistance felt as the rotor magnets snap past the stator teeth. Example: with the battery unplugged, turn a prop by hand and you feel small clicks — that is cogging, not damage.

MOTOR-022

Why are brushless motors preferred over brushed motors?

- A. Cheaper B. More Efficient ✓ C. Simpler D. Heavier

Why: Brushless motors are more efficient because there are no brushes causing friction and heat, and they also last longer and need less maintenance. Example: swapping brushed for brushless motors on a small build usually gives noticeably longer flight times.

MOTOR-023

Which motor typically has higher KV?

- A. 15-inch Drone Motor B. Tiny Whoop Motor ✓ C. Agricultural Drone Motor D. Heavy Lift Motor

Why: Tiny whoop motors spin small propellers on low voltage (often 1S), so they need very high KV to reach useful RPM. Large motors on 15-inch or heavy-lift drones use low KV. Example: a whoop motor may be around 19,000KV while a heavy-lift motor is about 100KV.

MOTOR-024**What does the first number in 2207 represent?**

- A. Stator Diameter ✓** B. Voltage C. RPM D. Prop Size

Why: In motor sizes like 2207 the first two digits are the stator diameter in millimetres (22mm); the last two are the stator height (07mm). Example: 2207 means a 22mm wide, 7mm tall stator.

MOTOR-025**Why is DC not supplied directly to brushless motor coils?**

- A. Motor would not continuously rotate ✓** B. Battery explodes C. ESC overheats D. Propeller falls off

Why: Feeding steady DC into a coil just holds the rotor locked to one magnetic position — it cannot keep rotating, and it can overheat the coil. The field must keep moving, so it has to be switched in sequence. Example: connecting DC to two phases makes the motor jump to one position and buzz.

MOTOR-026**Which motor characteristic mostly determines acceleration response?**

- A. Rotor Inertia ✓** B. Antenna Gain C. GPS Accuracy D. Receiver Protocol

Why: How quickly a motor speeds up or slows down is mostly set by its rotor inertia — the mass of the bell and magnets it must accelerate. Antenna gain, GPS and receiver protocol have nothing to do with it. Example: light racing motors react faster to throttle changes than heavy ones.

MOTOR-027**Tiny Whoops typically use:**

- A. Small Low-Power Motors ✓** B. Heavy Lift Motors C. Industrial Motors D. Car Motors

Why: Tiny whoops use very small, low-power motors (like 0802 or 1103 sizes) to suit their tiny frames and low-voltage batteries. Heavy-lift or industrial motors would be far too big. Example: a whoop motor weighs only a few grams.

MOTOR-028**A larger propeller usually requires:**

- A. Higher Torque ✓** B. Less Torque C. No Torque D. Smaller Motor

Why: A larger propeller pushes more air, so it takes more torque to spin it. Larger props therefore need a motor that produces higher torque. Example: 7-inch long-range builds use bigger, torquier motors than 5-inch freestyle quads.

MOTOR-029**What do the three motor wires represent?**

- A. Sensors B. Battery Lines **C. Three Electrical Phases ✓** D. Ground Connections

Why: The three motor wires carry the three electrical phases that the ESC energises in sequence to rotate the magnetic field. They are not sensors, battery lines or grounds. Example: the ESC pads for each motor are labelled A, B and C for the three phases.

MOTOR-030

Which motor is typically used on 5-inch freestyle drones?

- A. 0802 B. 1103 **C. 2207 ✓** D. 5008

Why: A 2207 motor is the common choice for 5-inch freestyle and racing quads. 0802 is for micro whoops, 1103 for small 3-inch builds and 5008 for large 10-inch long-range drones. Example: many 5-inch quads use 2207 motors around 1700–1950KV on 6S.

MOTOR-031

What does the second number in 2207 represent?

- A. KV **B. Stator Height ✓** C. Voltage D. RPM

Why: In 2207 the second pair of digits is stator height in millimetres (7mm). A taller stator holds more copper and magnet length, giving more torque. Example: a 2207 motor has a shorter stator than a 2307 or 2806.

MOTOR-032

What creates the magnetic field inside the motor?

- A. GPS **B. ESC Phase Switching ✓** C. Antenna D. Receiver

Why: The magnetic field is made by the ESC sending current through the stator windings in sequence — the coils become electromagnets. GPS, antennas and the receiver play no part. Example: as the ESC steps through its phase switching, the field appears to rotate and the rotor chases it.

MOTOR-033

Why do motor manufacturers use Neodymium magnets?

- A. Strong Magnetic Field B. Lightweight C. High Efficiency **D. All of the Above ✓**

Why: Neodymium magnets are very strong, light and efficient for their size, so all of those benefits apply. Example: small neodymium magnets inside the bell let a motor weighing about 30g lift well over a kilogram.

MOTOR-034

Why do racing drones prefer lightweight motors?

- A. Lower Rotational Inertia ✓** B. Better GPS C. Better Video D. More Weight

Why: Lightweight motors have lower rotational inertia, so they speed up and slow down faster — vital for racing's sharp throttle changes. Extra weight only slows response. Example: a racer punching out of a gate benefits from a light, quick-reacting motor.

MOTOR-035

A 15-inch heavy-lift drone motor commonly operates in which KV range?

- A. 2500–3000KV B. 1800–2200KV C. 800–1200KV **D. 100–400KV ✓**

Why: A 15-inch heavy-lift motor uses low KV (about 100–400KV) because it swings large props that need high torque at low RPM. The 1800–3000KV ranges are for small, fast props. Example: agricultural spray drones use motors around 100–300KV.

MOTOR-036

What component rotates in a brushless motor?

- A. Stator **B. Rotor ✓** C. ESC D. Bearing Housing

Why: The rotor is the part that rotates — in a typical FPV outrunner motor it is the outer bell with magnets attached to the shaft. The stator stays still. Example: when you spin a prop, the visible outer bell turns around the fixed core.

MOTOR-037

What is the relationship between motor current and torque?

- A. No Relationship **B. Torque Increases with Current ✓** C. Torque Decreases with Current D. Torque Equals Voltage

Why: Motor torque increases with current: more current through the windings creates a stronger magnetic force. Example: a hard throttle punch makes the motor draw a big current spike, giving high torque to accelerate the prop — and extra heat.

MOTOR-038

Lower KV motors generally provide:

- A. Higher Torque ✓** B. Lower Torque C. No Torque D. No Speed

Why: Lower-KV motors have more turns of thinner wire, producing stronger magnetic force per amp, so they give higher torque but lower top RPM. Example: a 1300KV motor swings a big prop with more torque than a 2300KV motor on the same pack.

MOTOR-039

What component remains stationary?

- A. Rotor B. Bell C. Shaft **D. Stator ✓**

Why: The stator is the stationary part, bolted to the motor base; the rotor, bell and shaft rotate. Example: the motor mounting screws hold the stator base to the frame arm while everything else spins.

MOTOR-040

A brushless motor is fundamentally:

- A. Transformer **B. Three-Phase Synchronous Motor ✓** C. Battery D. Receiver

Why: A brushless motor is fundamentally a three-phase synchronous motor: the rotor turns in step with the rotating magnetic field from the three phases. Example: if the ESC's timing is off or the load is too high the rotor loses sync, which is what a desync is.

MOTOR-041

Most FPV drones use:

- A. Brushed Motors **B. Brushless Motors ✓** C. Hydraulic Motors D. Stepper Motors

Why: Most FPV drones use brushless motors for their power, efficiency and reliability. Brushed motors only appear on very small or old builds; hydraulic and stepper motors are not used. Example: nearly every modern 5-inch quad runs four brushless motors.

MOTOR-042

Which winding material is commonly used in FPV motors?

- A. Aluminum B. Silver **C. Copper ✓** D. Gold

Why: Copper is the standard winding material because it conducts electricity very well at a reasonable price. Aluminium conducts worse, and silver and gold are far too expensive. Example: motor windings are enamelled copper wire.

MOTOR-043

Which part supports smooth motor rotation?

- A. Bearings ✓** B. Antenna C. Capacitor D. Diode

Why: Bearings let the shaft spin smoothly with very little friction. Antennas, capacitors and diodes do not support rotation. Example: a gritty, noisy motor after a crash usually means worn bearings that should be replaced.

MOTOR-044

How is motor direction reversed?

- A. Change Battery **B. Swap Phase Sequence ✓** C. Change Propeller D. Increase Voltage

Why: Swapping any two of the three phase wires reverses the phase sequence and so reverses the motor direction (or you can reverse it in the ESC software). Changing the battery, prop or voltage does not. Example: a new build with a motor spinning the wrong way is fixed by swapping two wires or toggling direction in the configurator.

MOTOR-045

Why do larger motors have greater torque?

- A. Larger Magnetic Radius B. More Copper Volume C. Stronger Magnetic Forces **D. All of the Above ✓**

Why: Larger motors have a larger magnetic radius (more leverage), more copper and stronger magnetic forces — all of which raise torque, so the answer is "all of the above". Example: a 2806 motor produces more torque than a 2204 to turn bigger props.

MOTOR-046

What happens if bearings fail?

- A. Reduced Efficiency B. More Friction C. Heat Generation **D. All of the Above ✓**

Why: Failing bearings cause more friction, which lowers efficiency and creates heat — all of the listed problems — and can eventually seize the motor. Example: a crunchy-feeling motor after a crash in sand should have its bearings replaced before the next flight.

MOTOR-047

What causes the rotor to follow the magnetic field?

- A. Gravity **B. Permanent Magnets ✓** C. GPS D. Air Pressure

Why: The rotor follows the rotating magnetic field because its permanent magnets are attracted to the stator's changing poles. Gravity, GPS and air pressure do not drive it. Example: as the ESC steps the field around, the magnets are pulled along behind it.

MOTOR-048

A 2000KV motor running on 10V ideally spins at:

- A. 200 RPM B. 2000 RPM **C. 20,000 RPM ✓** D. 200,000 RPM

Why: KV multiplied by voltage gives ideal no-load RPM: $2000\text{KV} \times 10\text{V} = 20,000\text{ RPM}$. Example: the same motor on a 4S pack (about 16.8V) would ideally reach about 33,600 RPM unloaded; with a propeller fitted real RPM is lower.

MOTOR-049

What is motor efficiency?

- A. Mechanical Power Output $\div$ Electrical Power Input ✓** B. RPM $\times$ Voltage C. Voltage $\div$ Current D. Current $\times$ Resistance

Why: Motor efficiency is mechanical power out divided by electrical power in. The rest is lost as heat. Example: 100W in and 85W of shaft power out means 85% efficiency, with 15W wasted as heat.

MOTOR-050

Why is AC-like three-phase power superior for brushless motors?

- A. Produces Smooth Rotating Magnetic Field ✓** B. Easier GPS Integration C. Reduces Battery Weight D. Improves Radio Range

Why: Three-phase power creates a smooth, continuously rotating magnetic field and steady torque, instead of the pulsing you would get from one phase. Example: smooth three-phase drive gives a quad a calmer hover and less vibration in the video.

2. Electronic Speed Controller (ESC) Mastery

How the ESC turns battery DC into three-phase drive: MOSFETs, DShot, current ratings, desync and protection parts.

50 questions

ESC-001

Tiny Whoops usually operate on:

- A. 1S–2S ✓** B. 6S–8S C. 10S–12S D. 14S–16S

Why: Tiny whoops are very small, light aircraft with tiny motors, so they run on 1S–2S batteries. 6S and above are for much bigger, higher-power drones. Example: a 65mm whoop flies on a single 1S 300mAh LiPo (about 3.7V).

ESC-002

MOSFET losses mainly generate:

- A. GPS Error **B. Heat ✓** C. Video Noise D. Vibration

Why: MOSFETs are not perfect switches, so some power is lost as heat whenever they conduct or switch. They do not cause GPS error, video noise or vibration. Example: an ESC running hard in a hot, airless frame gets hot enough that a heatsink or airflow helps.

ESC-003

Why are capacitors added near battery pads?

- A. Decoration **B. Noise Reduction ✓** C. Cooling D. Weight Balance

Why: Capacitors near the battery pads soak up voltage spikes and ripple caused by the ESC switching, which reduces electrical noise. Example: without the capacitor you may see lines or flicker in the FPV video when you punch the throttle.

ESC-004

Which formula calculates electrical power?

- A. $P = V \times I$ ✓** B. $P = I / V$ C. $P = V^2$ D. $P = R / I$

Why: Electrical power (in watts) is voltage multiplied by current: $P = V \times I$. It tells you how much energy a part is handling, which decides heat and the size of ESC and battery you need. Example: a motor drawing 30A from a 16V pack uses $16 \times 30 = 480W$, so four motors at full throttle can use nearly 2000W — which is why ESCs and wires get hot.

ESC-005

Sensorless ESCs determine rotor position using:

- A. GPS B. Gyroscope **C. Back EMF ✓** D. Magnetometer

Why: A sensorless ESC works out where the rotor is by reading the back EMF voltage the spinning motor generates. GPS, gyroscopes and magnetometers are not used for this. Example: the ESC watches back EMF to know exactly when to switch to the next phase.



ESC-006

What does MCU stand for?

- A. Main Control Unit **B. Microcontroller Unit ✓** C. Motor Control Utility D. Multi Current Unit

Why: MCU stands for Microcontroller Unit — the small computer chip that runs the ESC's firmware. Example: the MCU on a 4-in-1 ESC reads the DShot signals and decides how to switch the MOSFETs.

ESC-007

A larger propeller generally requires:

- A. Lower Torque **B. More Current ✓** C. Less Voltage D. Less Power

Why: A larger propeller moves more air and is harder to spin, so the motor needs more torque and therefore draws more current. Example: switching from a 5-inch to a 6-inch prop on the same motor raises current draw and can overheat the ESC.

ESC-008

What is telemetry?

- A. Video Signal **B. Data Feedback from ESC ✓** C. GPS Location D. Radio Frequency

Why: In ESC terms, telemetry is data the ESC sends back — such as current, temperature, voltage and RPM — rather than video, GPS or radio signals. Example: with ESC telemetry on, your OSD can show amps drawn and your flight controller logs motor RPM.

ESC-009

The purpose of current sensing is:

- A. Measure Motor Load ✓** B. Increase RPM C. Improve GPS D. Record Video

Why: Current sensing measures how much current the motors are drawing, i.e. their load. It does not change RPM, GPS or video. Example: the OSD "amps" readout and the battery mAh used both come from current sensing.

ESC-010

What component smooths voltage spikes?

- A. Capacitor ✓** B. Motor C. Antenna D. LED

Why: A capacitor stores and releases a little energy, which smooths voltage spikes. Motors, antennas and LEDs do not do this. Example: a low-ESR capacitor soldered to the battery pads keeps spikes from damaging the ESC.

ESC-011

The ESC converts battery DC power into:

- A. AC Power ✓** B. RF Signals C. GPS Data D. Video Signals

Why: The ESC takes the battery's DC and converts it into three-phase, AC-like drive for the brushless motor. It does not produce RF, GPS or video signals. Example: the three motor wires carry these switched, phase-shifted pulses.

ESC-012

Increasing PWM frequency generally:

- A. Makes ESC larger **B. Reduces Motor Noise ✓** C. Increases GPS Accuracy D. Reduces Voltage

Why: Raising the PWM switching frequency makes the current in the motor smoother, which generally reduces audible motor whine/noise (at the cost of a little more switching heat). Example: pilots raise ESC PWM frequency to cut motor noise and sometimes improve smoothness.

ESC-013

What component acts as an electronic switch?

- A. MOSFET ✓** B. Capacitor C. Resistor D. Coil

Why: A MOSFET is the electronic switch that turns current to the motor on and off. Capacitors, resistors and coils do not act as switches. Example: each motor phase uses MOSFETs to connect to the battery or ground thousands of times a second.

ESC-014

Which protocol is commonly used today?

- A. PWM **B. DShot600 ✓** C. UART D. SPI

Why: DShot600 is the common digital protocol between flight controller and ESC today, replacing older analog PWM. UART and SPI are general-purpose buses, not the ESC command protocol. Example: most modern quads are configured with DShot300 or DShot600.

ESC-015

Most FPV drones use what type of motor?

- A. Brushed DC Motor B. Stepper Motor **C. Brushless Motor ✓** D. Servo Motor

Why: Most FPV drones use brushless motors for efficiency, power and reliability. Brushed, stepper and servo motors are not used for propulsion. Example: a typical 5-inch freestyle quad has four brushless motors.

ESC-016

Which component protects against voltage spikes?

- A. TVS Diode ✓** B. Antenna C. GPS D. Magnet

Why: A TVS (transient voltage suppression) diode clamps dangerous voltage spikes. Antennas, GPS modules and magnets do not protect electronics from spikes. Example: a TVS diode on the battery lead helps protect the ESC when you plug in a battery.

ESC-017

DShot is:

- A. Analog Protocol **B. Digital Protocol ✓** C. Video Protocol D. RF Protocol

Why: DShot is a digital protocol: it sends motor commands as digital numbers, which are more precise and less noise-prone than analog. It is not a video or RF protocol. Example: DShot sends a value like 1047 meaning "motor at about half throttle".

ESC-018

What determines motor rotation direction?

- A. Battery Brand B. Propeller Color **C. Phase Sequence ✓** D. ESC Weight

Why: Motor direction is set by the order of the three phases (the phase sequence). Battery brand, prop colour and ESC weight have no effect. Example: swapping two motor wires reverses that sequence and the motor spins the other way.

ESC-019

A brushless motor's rotating magnetic field is produced by:

- A. ESC Phase Switching ✓** B. GPS Module C. Receiver D. Capacitor

Why: The rotating magnetic field comes from the ESC switching current through the phases in sequence. GPS modules, receivers and capacitors cannot make a field rotate. Example: the faster the ESC steps through the phases, the faster the rotor turns.

ESC-020

What is a gate driver?

- A. GPS Module B. Motor Bearing **C. MOSFET Driver Circuit ✓** D. Battery Charger

Why: A gate driver is the circuit that drives the MOSFET gates quickly and strongly so they switch cleanly. It has nothing to do with GPS, bearings or charging. Example: a slow or weak gate drive makes MOSFETs overheat because they spend too long half-on.

ESC-021

What does bidirectional DShot enable?

- A. GPS Navigation **B. RPM Feedback ✓** C. Video Transmission D. Battery Charging

Why: Bidirectional DShot lets the ESC send RPM data back to the flight controller over the same signal wire. It does not provide GPS, video or charging. Example: the flight controller uses that RPM feedback for RPM filtering, which cleans up gyro noise.

ESC-022

How many wires come out of a typical FPV brushless motor?

- A. 2 **B. 3 ✓** C. 4 D. 5

Why: A typical FPV brushless motor has three wires, one for each electrical phase (A, B and C). The ESC energises them in sequence to spin the motor; standard sensorless motors have no extra sensor wires. Example: a quad has four motors, so you solder 12 motor wires to the ESC pads, and swapping any two wires on a motor reverses its direction.

ESC-023

What is the purpose of a voltage regulator?

- A. Increase RPM **B. Reduce Voltage ✓** C. Improve GPS D. Generate Video

Why: A voltage regulator reduces (steps down) voltage to a safe lower level for electronics. It does not raise RPM, improve GPS or create video. Example: a regulator converts the battery's 16V down to 5V for the flight controller and receiver.

ESC-024

Which signal does a Flight Controller send to ESC?

- A. GPS Data **B. Motor Command ✓** C. Video Feed D. Telemetry Only

Why: The flight controller sends motor commands (how fast each motor should spin) to the ESC. GPS data and video feed do not go to the ESC. Example: when you push the throttle stick, the FC sends bigger motor command numbers to each ESC.

ESC-025

Which component acts as the brain of an ESC?

- A. MOSFET B. Capacitor **C. MCU ✓** D. Diode

Why: The MCU (microcontroller) is the brain of the ESC — it runs the firmware and decides how to switch the MOSFETs. MOSFETs, capacitors and diodes are components it controls or supports. Example: flashing BLHeli means loading new firmware onto the ESC's MCU.

ESC-026

What happens if two motor wires are swapped?

- A. ESC burns **B. Motor reverses direction ✓** C. Battery explodes D. RPM doubles

Why: Swapping two motor wires reverses the phase sequence, so the motor spins in the opposite direction. It does not harm the ESC or double the RPM. Example: if a motor spins the wrong way on a new build, swap any two of its wires.

ESC-027

Which firmware is popular in FPV ESCs?

- A. Windows B. Android **C. BLHeli ✓** D. Ubuntu

Why: BLHeli is popular ESC firmware for FPV. Windows, Android and Ubuntu are operating systems, not ESC firmware. Example: pilots flash BLHeli_S or BLHeli_32 onto their ESCs using a configurator.

ESC-028

What is ESC desync?

- A. Battery Failure **B. Rotor Position Tracking Failure ✓** C. GPS Drift D. Propeller Damage

Why: Desync happens when the ESC loses track of the rotor's position, so its switching no longer matches the rotor — the motor stutters or stops. It is not a battery, GPS or prop issue. Example: a motor suddenly stuttering and a quad tumbling mid-air after a hard throttle punch is often a desync.

ESC-029

What does ESC stand for?

- A. Electronic Speed Controller ✓** B. Electronic Signal Converter C. Engine Speed Controller D. Electric Switching Circuit

Why: ESC stands for Electronic Speed Controller. It receives the throttle command from the flight controller and turns the battery's DC power into the three-phase drive that makes the brushless motor spin at exactly that speed. Example: when you push the throttle stick, the flight controller sends a bigger number to the ESC and the ESC drives the motor faster.



ESC-030

Which motor KV is generally used for heavy-lift 15-inch drones?

- A. 3000 KV B. 2500 KV C. 1500 KV **D. 100–400 KV ✓**

Why: A 15-inch heavy-lift drone uses low-KV motors, around 100–400KV, to turn large props with high torque. 1500–3000KV is for small, fast props. Example: agricultural drones use motors near 100–300KV.

ESC-031

A 5-inch freestyle drone typically uses:

- A. 5A ESC B. 15A ESC **C. 45A ESC ✓** D. 200A ESC

Why: A 5-inch freestyle drone usually pairs with a 45A-class ESC to supply the motors' peak current. 5A or 15A would be overloaded and 200A is oversized for this build. Example: many 5-inch quads use a 45A or 55A 4-in-1 ESC.

ESC-032

Which drone typically needs the highest current ESC?

- A. Tiny Whoop B. 3-inch C. 5-inch **D. 15-inch Heavy Lift ✓**

Why: The biggest, heaviest drones with large props and motors draw the highest current, so a 15-inch heavy-lift drone needs the highest-current ESC. Tiny whoops draw only a few amps. Example: a 15-inch lifter's ESCs may need 80A or more.

ESC-033

How many MOSFETs are generally required for one brushless motor?

- A. 2 B. 4 **C. 6 ✓** D. 12

Why: A three-phase motor needs six MOSFETs: a high-side and a low-side switch for each of the three phases ($3 \times 2 = 6$). Example: this is the standard "H-bridge per phase" layout used in a brushless ESC.

ESC-034

What is Back EMF?

- A. Battery Voltage **B. Motor-generated Voltage ✓** C. ESC Output Voltage D. Receiver Signal

Why: Back EMF is the voltage the motor itself generates as its magnets spin past the windings. It is not battery voltage, ESC output or a receiver signal. Example: a spinning motor can act as a small generator and produce a measurable voltage.

ESC-035

Current is measured in:

- A. Volts B. Watts **C. Amps ✓** D. Ohms

Why: Current is measured in amps (A). Volts measure electrical pressure, watts measure power and ohms measure resistance. Amps matter because they decide how hot wires, MOSFETs and motors get. Example: a 45A ESC is rated to handle 45 amps continuously; if your motors pull 50A for a long time the ESC will overheat.



ESC-036

A 4-in-1 ESC for a quadcopter typically contains:

- A. 12 MOSFETs **B. 24 MOSFETs ✓** C. 48 MOSFETs D. 96 MOSFETs

Why: Each motor needs its own driver of six MOSFETs: a high-side and a low-side switch for each of the three phases ($3 \times 2 = 6$). A 4-in-1 ESC drives four motors, so it holds $4 \times 6 = 24$ MOSFETs. Example: if one MOSFET burns out, usually only the motor connected to that section loses power, not the whole ESC board.

ESC-037

A 4-in-1 ESC controls how many motors?

- A. 1 B. 2 **C. 4 ✓** D. 8

Why: A 4-in-1 ESC contains four separate motor drivers on one board, so it controls four motors, one for each corner of a quadcopter. Example: using a single 4-in-1 board instead of four separate ESCs saves weight and wiring and makes the build much neater; a hexacopter with six motors would need more ESC outputs.

ESC-038

The ESC receives power directly from:

- A. Flight Controller B. Receiver **C. Battery ✓** D. GPS

Why: The ESC takes its power directly from the battery, since it must handle the large motor currents. The flight controller, receiver and GPS draw only small power from a regulator. Example: the thick battery leads connect straight to the ESC.

ESC-039

Which component is most likely to fail first during extreme overcurrent?

- A. Antenna **B. MOSFET ✓** C. Receiver D. GPS

Why: When there is too much current, the MOSFETs overheat or fail first because they carry all the motor current. Antennas, receivers and GPS are low-power parts. Example: a stalled motor can pop a MOSFET and release magic smoke from the ESC.

ESC-040

What occurs during MOSFET shoot-through?

- A. Motor Stops **B. Direct Battery Short ✓** C. RPM Increase D. Signal Loss

Why: MOSFET shoot-through is when the high-side and low-side MOSFETs in the same leg are on at the same time, creating a direct short across the battery. It does not simply stop the motor. Example: it can instantly destroy an ESC.

ESC-041

What does "40A ESC" indicate?

- A. Weight B. Voltage **C. Maximum Continuous Current ✓** D. RPM

Why: A "40A ESC" is rated for 40 amps of maximum continuous current. It does not indicate weight, voltage or RPM. Example: pairing a 40A ESC with motors that draw 50A continuously will overheat it.

ESC-042

A 6S battery has a nominal voltage of approximately:

- A. 7.4V B. 11.1V **C. 22.2V ✓** D. 44.4V

Why: Each LiPo cell is about 3.7V nominal, so a 6S pack (six cells in series) is $6 \times 3.7 = 22.2\text{V}$ nominal. A fully charged cell is 4.2V, so a full 6S pack reads about 25.2V. Example: 2S is 7.4V, 3S is 11.1V and 12S is 44.4V, so those answers belong to other pack sizes.

ESC-043

Why do ESCs need dead-time control?

- A. Prevent GPS Loss **B. Avoid MOSFET Shoot-Through ✓** C. Improve Video Quality D. Increase Range

Why: Dead-time is a tiny delay between turning one MOSFET off and the other on, so they are never on together and cannot cause shoot-through. It has nothing to do with GPS, video or range. Example: without dead-time the ESC could short the battery.

ESC-044

The primary function of the ESC in an FPV drone is:

- A. Generate Video B. Calculate GPS Coordinates **C. Convert DC Battery Power into Controlled 3-Phase AC for Brushless Motors ✓** D. Control Radio Communication

Why: The ESC's primary job is to convert the battery's DC power into controlled three-phase AC-like drive for the brushless motors. Video, GPS and radio are handled by other parts. Example: the flight controller says "spin faster" and the ESC changes its three-phase switching to match.

ESC-045

A 15-inch heavy lift drone running 12S batteries most likely requires:

- A. 5A ESC B. 10A ESC C. 15A ESC **D. 80A+ ESC ✓**

Why: A 15-inch heavy-lift drone on 12S draws high current to turn its large props, so it needs an ESC rated 80A or more. 5A–15A ESCs would burn out. Example: heavy-lift builds commonly use 80A–120A ESCs.

ESC-046

A 12S battery pack contains how many LiPo cells?

- A. 6 B. 8 C. 10 **D. 12 ✓**

Why: The "S" number is how many cells are connected in series, so a 12S pack has 12 LiPo cells, giving about $12 \times 3.7 = 44.4\text{V}$ nominal (50.4V when full). Example: heavy-lift drones use 12S because higher voltage delivers the same power with less current, allowing thinner wires and cooler ESCs.

ESC-047

What does RPM mean?

- A. Rotations Per Minute ✓** B. Rotor Pulse Management C. Rapid Power Module D. Resistance Per Motor

Why: RPM means Rotations Per Minute, how many full turns the motor makes each minute. Motor speed decides thrust, noise and how hard the ESC works. Example: a 2000KV motor on 10V ideally spins about 20,000 RPM, and with bidirectional DShot the ESC reports the real RPM back to the flight controller so it can filter motor noise out of the gyro.



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ESC-048

What is the main job of an ESC?

- A. Store battery power **B. Control motor speed ✓** C. Receive GPS signals D. Record video

Why: The ESC's main job is to control motor speed based on commands from the flight controller. It does not store power, receive GPS or record video. Example: more throttle means the ESC drives its motor faster.

ESC-049

Which ESC firmware supports bidirectional DShot?

- A. DOS **B. BLHeli32 ✓** C. Android D. Linux

Why: BLHeli_32 supports bidirectional DShot (RPM feedback). DOS, Android and Linux are operating systems, not ESC firmware. Example: with BLHeli_32 ESCs you can enable bidirectional DShot to run RPM filtering.

ESC-050

Which battery is commonly used in FPV drones?

- A. Lead Acid B. NiCd **C. LiPo ✓** D. Alkaline

Why: LiPo is the common FPV battery thanks to its high energy density and ability to deliver big currents. Lead acid is too heavy, and NiCd and alkaline cannot supply such current. Example: nearly every 5-inch quad flies on a 4S or 6S LiPo.

3. Flight Controller (FC) Architecture

The drone's brain: gyro and sensors, PID control, motor mixing, flight modes, firmware and Blackbox.

50 questions

FLTCTRL-001

What happens inside the FC after receiving stick input?

- A. PID Calculations B. Sensor Fusion C. Motor Mix Computation **D. All of the Above ✓**

Why: After a stick input the flight controller runs all of these steps: it fuses the sensor data, runs the PID calculations against what the pilot asked for, and computes the motor mix — so "all of the above". Example: push the right stick forward and the FC works out exactly how much faster the rear motors must spin and by how much slower the front ones.

FLTCTRL-002

Which flight mode self-levels the drone?

- A. Acro B. Manual **C. Angle Mode ✓** D. Race Mode

Why: Angle mode uses the accelerometer to self-level: let go of the sticks and the drone returns to level. Acro, manual and race modes only hold a rotation rate and never self-level. Example: beginners fly in Angle mode so the quad stays upright even when they make mistakes.

FLTCTRL-003

What is the role of the motor mixer in an X-Quad?

- A. Calculate Individual Motor Throttle Outputs ✓** B. Control Battery Voltage C. Generate GPS Data D. Record Video

Why: The motor mixer turns the pilot's roll, pitch, yaw and throttle demands into an individual throttle value for each of the four motors. It has nothing to do with battery voltage, GPS or video. Example: for a roll right, it speeds up the left motors and slows the right ones.

FLTCTRL-004

What causes oscillations in drones?

- A. Poor PID Tuning ✓** B. Weak GPS C. Antenna Placement D. OSD Settings

Why: Oscillations (the quad wobbling or shaking) are usually caused by poor PID tuning, such as a P gain that is too high or a D gain that is too low. Weak GPS, antenna placement or OSD settings do not cause oscillation. Example: a quad that shakes rapidly after punching the throttle needs its PIDs or filters adjusted.

FLTCTRL-005

A magnetometer measures:

- A. Current B. Temperature **C. Magnetic Direction ✓** D. RPM

Why: A magnetometer (compass) measures the direction of the Earth's magnetic field, so the drone knows which way it is facing. It does not measure current, temperature or RPM. Example: GPS return-to-home uses the compass heading to point the drone back toward you.

FLTCTRL-006

What is motor mixing?

- A. Converting Desired Motion into Individual Motor Commands ✓** B. Mixing Motor Oil C. Combining Batteries D. Changing Props

Why: Motor mixing means converting the desired motion (roll, pitch, yaw, thrust) into separate commands for each motor. It has nothing to do with motor oil, batteries or props. Example: to yaw, the mixer speeds up the clockwise pair and slows the counter-clockwise pair.

FLTCTRL-007

Which FC processor is generally more powerful?

- A. F4 **B. F7 ✓** C. F1 D. F3

Why: F7 processors are more powerful than F4, F1 or F3 ones, with a faster core and more resources. Example: flight controllers with an F7 chip can run faster gyro and PID loops and handle more features at once.

FLTCTRL-008

What is sensor fusion?

- A. Combining Multiple Sensors for Better Estimation ✓** B. Combining Batteries C. Combining ESCs D. Combining Motors

Why: Sensor fusion combines data from several sensors (like the gyro, accelerometer and barometer) to get a better estimate of the drone's state than any one sensor gives alone. It is not about combining batteries, ESCs or motors. Example: gyro and accelerometer data are blended to work out tilt accurately.

FLTCTRL-009

The Flight Controller is often called the:

- A. Motor B. ESC **C. Brain of the Drone ✓** D. Receiver

Why: The flight controller is often called the brain of the drone because it processes the sensors and pilot commands and decides what every motor does. The motors, ESCs and receiver are parts it controls or listens to. Example: it keeps a quad hovering by constantly correcting the motors.

FLTCTRL-010

Why are F7 processors preferred over F4 in modern FCs?

- A. Faster Processing B. More UARTs C. Better Peripheral Support **D. All of the Above ✓**

Why: F7 processors are faster, have more UARTs and support more peripherals than F4 ones, so all of those reasons apply. Example: an F7 board can offer more UART ports for GPS, receiver and VTX at the same time.

FLTCTRL-011

What is the primary sensor used for stabilization?

- A. Gyroscope ✓** B. GPS C. Compass D. Camera

Why: The gyroscope is the main stabilisation sensor: it measures how fast the drone is rotating, and the FC corrects that. GPS, compass and the camera are not used for stabilisation. Example: when a gust tips the quad, the gyro detects the rotation and the FC corrects it within milliseconds.

FLTCTRL-012

Which firmware is most commonly used on FPV Flight Controllers?

- A. Windows B. Android **C. Betaflight ✓** D. Ubuntu

Why: Betaflight is the most widely used FPV flight controller firmware. Windows, Android and Ubuntu are operating systems. Example: you flash Betaflight onto the FC and tune it in the Betaflight Configurator.

FLTCTRL-013

What is UART?

- A. Sensor **B. Serial Communication Port ✓** C. Battery Type D. Firmware

Why: UART is a serial communication port that lets the FC talk to devices like receivers, GPS modules and VTXs. It is not a sensor, battery type or firmware. Example: your receiver connects to one UART and your GPS to another.

FLTCTRL-014

What is a barometer used for?

- A. RPM Measurement **B. Altitude Estimation ✓** C. Video Processing D. Radio Control

Why: A barometer measures air pressure, which the FC uses to estimate altitude. It does not measure RPM, process video or handle radio control. Example: altitude-hold modes use the barometer to keep the drone at a steady height.

FLTCTRL-015

What communication protocol is commonly used between FC and ESC?

- A. HDMI B. Ethernet **C. DShot ✓** D. USB

Why: DShot is the digital protocol commonly used between the FC and the ESC. HDMI, Ethernet and USB are not used for that. Example: the FC sends DShot motor commands down a single signal wire to each ESC.

FLTCTRL-016

What does FC stand for?

- A. Flight Controller ✓** B. Flight Computer C. Flying Circuit D. Flight Chip

Why: FC stands for Flight Controller, the circuit board that is the drone's brain. It carries the main processor, the gyro and accelerometer, and connections for the receiver, ESC and other parts, and it runs the firmware that keeps the drone flying. Example: in a stack it sits between the ESC and the VTX, and Betaflight runs on it.

FLTCTRL-017

What is the purpose of a BEC?

- A. Reduce Voltage for Electronics ✓** B. Increase RPM C. GPS Tracking D. Video Recording

Why: A BEC (battery eliminator circuit) steps the battery voltage down to a lower, steady voltage (such as 5V) to power electronics. It does not increase RPM or track GPS. Example: the BEC powers the receiver and GPS from the main battery.



FLTCTRL-018

What is the primary purpose of a Flight Controller?

- A. Transmit Video **B. Control Drone Stability ✓** C. Charge Battery D. Generate GPS Signals

Why: The flight controller's main purpose is to control the drone's stability (and flight) by constantly reading sensors and adjusting the motors. It does not transmit video, charge batteries or make GPS signals. Example: it holds a steady hover even in light wind.

FLTCTRL-019

The FC sends motor commands to:

- A. GPS B. Receiver **C. ESC ✓** D. Camera

Why: The FC sends motor commands to the ESCs, which actually drive the motors. It does not command the GPS, receiver or camera. Example: when you move the throttle stick, new motor values go from the FC to each ESC.

FLTCTRL-020

What sensor measures rotational movement?

- A. GPS **B. Gyroscope ✓** C. Barometer D. Compass

Why: A gyroscope measures rotational movement (how fast the drone is turning). GPS, barometers and compasses measure other things. Example: if the drone rolls to the right, the gyro reports the roll rate.

FLTCTRL-021

The Flight Controller receives gyro data typically at:

- A. Thousands of Times per Second ✓** B. Once per Minute C. Once per Second D. Once per Hour

Why: Flight controllers read the gyroscope thousands of times per second (for example 8kHz) so they can react almost instantly. Reading once per second or slower would be far too slow to stabilise a quad. Example: an 8kHz gyro is read every 125 microseconds.

FLTCTRL-022

What happens if the gyroscope fails?

- A. Drone Loses Stability ✓** B. GPS Stops C. Battery Drains D. ESC Burns

Why: If the gyroscope fails, the FC no longer knows how the drone is rotating, so the drone loses stability. GPS, battery and ESCs are not directly affected. Example: a damaged gyro makes the quad flip over or spin uncontrollably at takeoff.

FLTCTRL-023

A gyroscope measures:

- A. Position **B. Rotation Rate ✓** C. Altitude D. Voltage

Why: A gyroscope measures rotation rate (angular velocity). Position, altitude and voltage come from other sensors. Example: it reports something like 200 degrees per second when you roll.

FLTCTRL-024

GPS modules commonly connect using:

- A. UART ✓** B. HDMI C. USB D. PWM

Why: GPS modules commonly connect to the flight controller over a UART serial port. HDMI, USB and PWM are not used for this. Example: a GPS plugs into the FC's UART pads with TX/RX wires.

FLTCTRL-025

Why is soft mounting often used for FCs?

- A. Reduce Vibration Noise ✓** B. Increase Voltage C. Improve GPS D. Improve Video Resolution

Why: Soft mounting (rubber grommets) isolates the FC from motor vibration, reducing noise in the gyro readings. It does not raise voltage or improve GPS or video. Example: without soft mounting, vibrations make the quad twitch or the motors run hot.

FLTCTRL-026

Which processor is commonly used in modern FPV FCs?

- A. STM32F405 ✓** B. Pentium C. Ryzen D. Xeon

Why: STM32F405 is a typical microcontroller in modern FPV flight controllers. Pentium, Ryzen and Xeon are desktop and server CPUs. Example: many popular FCs are built around an STM32F405 chip.

FLTCTRL-027

Which component stores FC firmware?

- A. Flash Memory ✓** B. Motor C. Capacitor D. Antenna

Why: Firmware is stored in flash memory, which keeps data when power is off. Motors, capacitors and antennas do not store firmware. Example: flashing Betaflight writes the firmware into the MCU's flash memory.

FLTCTRL-028

The FC receives commands from:

- A. Receiver ✓** B. Motor C. ESC D. Battery

Why: The FC receives the pilot's commands from the receiver. Motors, ESCs and the battery do not send commands to it. Example: stick movements travel from the transmitter to the receiver and then to the FC.

FLTCTRL-029

Which software is used to configure Betaflight FCs?

- A. VS Code **B. Betaflight Configurator ✓** C. Photoshop D. KiCad

Why: Betaflight FCs are configured with the Betaflight Configurator. VS Code, Photoshop and KiCad are for coding, images and circuit design. Example: you plug the FC in by USB and set up receiver, modes and PIDs in the Configurator.

FLTCTRL-030**Blackbox logging is used for:**

- A. Flight Analysis ✓** B. Charging Battery C. GPS Navigation D. Radio Communication

Why: Blackbox logging records flight data so you can analyse a flight afterwards, for example to tune PIDs or find noise. It does not charge batteries or provide navigation. Example: after a flight with wobbles, you review the blackbox log to see what happened.

FLTCTRL-031**Which mode is preferred by FPV racers?**

- A. Angle B. Horizon **C. Acro ✓** D. GPS Mode

Why: Racers prefer Acro mode because it gives full manual control with no self-levelling, allowing fast, precise manoeuvres. Angle, Horizon and GPS modes limit or assist the aircraft. Example: in Acro, the quad holds its tilt until you move the stick back.

FLTCTRL-032**What best describes a Flight Controller?**

- A. Real-Time Embedded Computer for Drone Stabilization and Navigation ✓** B. Battery Charger C. Video Recorder D. Radio Transmitter

Why: The flight controller is a real-time embedded computer that stabilises and navigates the drone. It is not a charger, video recorder or radio transmitter. Example: it runs sensor and motor loops thousands of times each second.

FLTCTRL-033**What is ICM42688?**

- A. Gyroscope Chip ✓** B. ESC C. GPS Module D. Receiver

Why: The ICM42688 is a gyroscope (IMU) chip commonly found on modern flight controllers. It is not an ESC, GPS module or receiver. Example: an FC with an ICM42688 offers high-rate, low-noise gyro data.

FLTCTRL-034**What is the purpose of USB on a Flight Controller?**

- A. Connect to Configurator ✓** B. Increase Speed C. Power Motors D. GPS Tracking

Why: The USB port on a flight controller is used to connect to the configurator on a computer. It does not power the motors or provide GPS. Example: you plug USB into the FC to change settings and flash firmware.

FLTCTRL-035**Why are low-latency control loops important?**

- A. Faster Drone Response ✓** B. Better Charging C. More GPS Satellites D. More Video Quality

Why: Low-latency control loops make the drone respond faster to stick inputs and disturbances. They do not improve charging, GPS or video quality. Example: faster gyro and PID loops make the quad feel more locked-in.

FLTCTRL-036**How does the FC know the drone is tilting?**

- A. Gyroscope Data ✓**
- B. ESC Current C. Battery Voltage D. Camera Feed

Why: The FC knows the drone is tilting from the gyroscope data (assisted by the accelerometer). ESC current, battery voltage and camera feed do not report tilt. Example: a gust rolls the quad and the gyro immediately reports it.

FLTCTRL-037**What is the role of DMA in FC processors?**

- A. Faster Data Transfer ✓**
- B. Battery Monitoring C. GPS Processing D. Video Encoding

Why: DMA (direct memory access) lets peripherals move data without using the processor for every byte, giving faster data transfer. It is not for battery monitoring, GPS or video encoding. Example: DMA helps send fast ESC protocols without slowing the main loop.

FLTCTRL-038**What does OSD stand for?**

- A. On Screen Display ✓**
- B. Output Signal Driver C. Optical Signal Device D. Open Source Display

Why: OSD stands for On Screen Display — the information overlaid on your FPV video such as battery voltage and flight mode. Example: the OSD shows your battery voltage and timer in the goggles.

FLTCTRL-039**What voltage commonly powers the FC MCU?**

- A. 48V B. 22V
- C. 3.3V ✓**
- D. 120V

Why: The flight controller's MCU is a low-voltage chip that runs on 3.3V logic. The board's regulators step the battery down to 5V and then to 3.3V for it, which is why you never connect battery voltage directly to a signal pad. Example: a short or wrong wire from the battery to a sensor pad can instantly destroy the processor.

FLTCTRL-040**What is PID in drone control?**

- A. Position Integration Device
- B. Proportional Integral Derivative ✓**
- C. Power Input Driver D. Pulse Identification Decoder

Why: PID stands for Proportional, Integral, Derivative — the three terms of the control algorithm that stabilises the drone. Example: P reacts to current error, I corrects drift and D damps fast changes.

FLTCTRL-041**Which company manufactures many FC processors?**

- A. NVIDIA
- B. STMicroelectronics ✓**
- C. Sony D. Intel

Why: STMicroelectronics manufactures the STM32 processors used in many FCs. NVIDIA, Sony and Intel do not make these. Example: an STM32F405 or F722 is made by STMicroelectronics.



FLTCTRL-042

The OSD displays information on:

- A. Motors **B. Video Feed ✓** C. Battery D. ESC

Why: The OSD displays information on the video feed that you see in your goggles. It does not display on motors, batteries or ESCs. Example: battery voltage appears as text over the live video.

FLTCTRL-043

What does Blackbox record?

- A. Sensor Data ✓** B. GPS Only C. Battery Only D. Video Only

Why: Blackbox records sensor data while you fly: gyro and accelerometer readings, stick inputs, motor outputs and PID values, many times per second, to onboard flash or an SD card. It is a flight recorder, not just GPS, battery or video. Example: if your quad oscillates, you open the log in Blackbox Explorer, see where the noise comes from and adjust the filters or PIDs.

FLTCTRL-044

In a modern FPV drone, the Flight Controller primarily performs:

- A. Real-Time Sensor Processing B. PID Computation C. Motor Command Generation **D. All of the Above ✓**

Why: The flight controller continuously does all three jobs: it reads the sensors in real time, runs the PID calculation to compare what you want with what the drone is doing, and generates the motor commands. Example: when you pitch forward, it reads the gyro, works out the error, and sends new speeds to the four ESCs within a fraction of a millisecond, thousands of times per second.

FLTCTRL-045

Which component acts as the brain of the FC?

- A. MCU ✓** B. MOSFET C. Capacitor D. Diode

Why: The MCU (microcontroller) is the brain of the FC, running the firmware. MOSFETs, capacitors and diodes are supporting components. Example: the STM32 chip on the board is the MCU.

FLTCTRL-046

An accelerometer measures:

- A. Temperature B. Speed **C. Acceleration ✓** D. RPM

Why: An accelerometer measures acceleration (including gravity), which helps estimate tilt. It does not measure temperature, speed or RPM directly. Example: when still, it measures gravity and shows which way is down.

FLTCTRL-047

Which equation best represents the PID controller?

- A. $P + I + D$ ✓** B. $V \times I$ C. $F = ma$ D. $P = VI$

Why: The PID controller adds its three terms: output = $P + I + D$. P reacts to the current error, I builds up error over time to remove slow drift, and D reacts to how fast the error is changing to damp overshoot. Example: P too high makes fast shaking, D too low gives bounce-back after a flip, and I too low lets the quad drift in wind.

FLTCTRL-048**What is the purpose of PID control?**

- A. Stabilize Drone Motion ✓** B. Increase Battery Capacity C. Improve Video D. Increase GPS Accuracy

Why: PID control is used to stabilise the drone's motion by constantly correcting error between the desired and actual motion. It does not increase battery capacity, video quality or GPS accuracy. Example: it keeps the quad level and responsive to sticks.

FLTCTRL-049**What is the FC's main control loop frequency?**

- A. Sensor Reading + PID Computation Cycle ✓** B. Battery Charge Cycle C. GPS Refresh Rate D. Motor RPM

Why: The FC's main control loop is the cycle of sensor reading plus PID computation, repeated very fast (for example 8kHz). It is not the battery charge cycle, GPS refresh or motor RPM. Example: a faster loop means quicker corrections.

FLTCTRL-050**What sensor measures acceleration?**

- A. Accelerometer ✓** B. GPS C. Magnetometer D. ESC

Why: An accelerometer measures acceleration, including the constant pull of gravity, which lets the flight controller tell which way is down. The gyro measures rotation rate instead. Example: in Angle mode the flight controller uses the accelerometer to know when the quad is level and holds it there; GPS, magnetometers and ESCs do not measure acceleration.

4. Propeller Aerodynamics

Thrust, pitch, airfoils, Bernoulli and Newton, prop wash, disk loading and how blade count and size change flight.

50 questions

PROP-001

Why is propeller balance important?

- A. Reduce Vibrations B. Improve Efficiency C. Protect Bearings **D. All of the Above ✓**

Why: An unbalanced propeller makes the motor vibrate, wastes energy and wears bearings — so balance reduces vibration, improves efficiency and protects the bearings. Example: a slightly heavy blade shows up as jello in the video and a motor that runs hot.

PROP-002

What creates lift/thrust on a propeller?

- A. Airflow over blades ✓** B. Battery Voltage C. ESC Temperature D. GPS Signal

Why: Thrust comes from the airflow over the spinning blades: their shape and angle accelerate air downward. Battery voltage, ESC temperature and GPS do not create thrust. Example: spin a prop in still air and you can feel a strong stream of air pushed down.

PROP-003

A bent propeller usually causes:

- A. Vibrations B. Reduced Efficiency C. Poor Flight Performance **D. All of the Above ✓**

Why: A bent propeller is unbalanced and has the wrong shape, causing vibration, lower efficiency and poor flight performance — all of the above. Example: after hitting a branch, a bent prop gives visible vibration and the quad feels loose.

PROP-004

According to Bernoulli's Principle, lift increases when:

- A. Air Velocity Increases ✓** B. Air Velocity Decreases C. Voltage Increases D. RPM Decreases

Why: Bernoulli's principle says that where air moves faster, pressure is lower. Faster airflow over the blade top lowers pressure there, so lift increases. Slower air, voltage or lower RPM do not do this. Example: faster-spinning blades create more pressure difference and more lift.

PROP-005

Lower disk loading generally results in:

- A. Better Efficiency B. Longer Flight Time C. Lower Power Consumption **D. All of the Above ✓**

Why: Lower disk loading (less weight per propeller area) gives better efficiency, longer flight time and lower power use — all together. Example: a lightly loaded drone with large props hovers on much less power.

PROP-006

What principle explains thrust generation?

- A. Newton's Third Law ✓** B. Ohm's Law C. Faraday's Law D. Coulomb's Law

Why: Thrust is explained by Newton's Third Law: the propeller pushes air down, and the air pushes the drone up. Ohm's, Faraday's and Coulomb's laws are about electricity. Example: a prop accelerating air downward lifts the quad.

PROP-007

Why do heavy-lift drones use low-KV motors and large props?

- A. Produce High Torque Efficiently B. Move More Air C. Improve Flight Time **D. All of the Above ✓**

Why: Heavy-lift drones use low-KV motors and large props because they produce high torque efficiently, move a lot of air and improve flight time — so all of the above. Example: a 15-inch lifter with 100–400KV motors carries payloads efficiently.

PROP-008

What is propeller pitch?

- A. Blade Color **B. Distance Prop Moves in One Rotation (Ideal) ✓** C. Weight D. Thickness

Why: Propeller pitch is the theoretical distance the propeller would move forward in one full rotation. Colour, weight and thickness are not pitch. Example: a 5x4.3 prop has a pitch of 4.3 inches per revolution in ideal conditions.

PROP-009

What happens if blade count increases?

- A. More Grip in Air B. More Current Draw C. More Thrust **D. All of the Above ✓**

Why: More blades mean more grip in the air, more thrust and more current draw, so all of the options happen. Example: a tri-blade prop gives more bite than a bi-blade but needs more current.

PROP-010

A propeller can be viewed as:

- A. Multiple Rotating Wings ✓** B. Battery Pack C. ESC D. Radio System

Why: A propeller can be viewed as multiple small rotating wings, each blade working like a wing generating lift. It is not a battery, ESC or radio. Example: each blade has an airfoil cross-section just like a plane's wing.

PROP-011

What happens if angle of attack becomes too large?

- A. Stall Occurs ✓** B. More Lift Forever C. More Efficiency D. Zero Drag

Why: If the angle of attack gets too large, airflow separates from the blade and stalls, so lift drops and drag rises. More lift forever, more efficiency or zero drag do not happen. Example: a propeller that is overloaded or in turbulent air loses thrust when it stalls.

PROP-012

According to Newton's Third Law, propellers create thrust by:

- A. Accelerating Air Backward ✓** B. Accelerating Air Forward C. Reducing Voltage D. Increasing Current

Why: By Newton's Third Law, a propeller creates thrust by accelerating air backward (downward in a hover), which pushes the drone the other way. Accelerating air forward would push it backward. Example: the prop pushes air down and the quad rises.

PROP-013

A propeller converts:

- A. Heat Energy to Sound **B. Rotational Energy into Thrust ✓** C. GPS into Motion D. Voltage into Current

Why: A propeller converts the motor's rotational energy into thrust (a pushing force). It does not convert GPS into motion or heat into sound. Example: the motor's spin becomes lift through the blades.

PROP-014

Which prop generally provides more thrust?

- A. Larger Diameter ✓** B. Smaller Diameter C. Broken Prop D. Bent Prop

Why: A larger-diameter propeller moves more air and generally gives more thrust. Smaller, broken or bent props give less. Example: swapping a 5-inch for a 6-inch prop typically increases thrust (and current draw).

PROP-015

What is disk loading?

- A. Weight Supported per Propeller Area ✓** B. Battery Capacity C. ESC Current D. Motor Voltage

Why: Disk loading is the weight supported per unit of propeller area. It is not battery capacity, ESC current or motor voltage. Example: a heavy quad with small props has high disk loading; a light one with big props has low disk loading.

PROP-016

What is blade loading?

- A. Force Acting on Blade Area ✓** B. Battery Load C. ESC Load D. GPS Load

Why: Blade loading is the force (thrust) carried by the blade's area. Disk loading looks at the whole rotor area instead. High blade loading makes blades flex or stall and lowers efficiency. Example: a small-bladed prop on a heavy quad has high blade loading and loses grip sooner during hard throttle punches.

PROP-017

What creates high pressure below the blade?

- A. Slower Airflow ✓** B. Antenna C. Magnet D. Receiver

Why: Slower airflow under the blade means higher pressure there. Antennas, magnets and receivers do not create pressure. Example: pressure is higher below the blade and lower above, which pushes the blade upward.

PROP-018

How many blades does a typical FPV racing prop have?

- A. 1 B. 2 **C. 3 ✓** D. 6

Why: Most FPV racing props have three blades, a good compromise between thrust and efficiency. Single or six-blade props are uncommon. Example: popular 5-inch racing props are tri-blade.

PROP-019

Which prop type is most common in freestyle FPV?

- A. Tri-blade ✓** B. Single Blade C. Helicopter Blade D. Six Blade

Why: Tri-blade props are the most common for freestyle, giving good thrust and control. Single, helicopter and six-blade props are rare. Example: most freestyle quads fly tri-blade 5-inch props.

PROP-020

Which side of an airfoil typically has lower pressure?

- A. Top Surface ✓** B. Bottom Surface C. Hub D. Shaft

Why: The top surface of an airfoil has lower pressure because the air moves faster over it. The bottom has higher pressure; the hub and shaft are not airfoil surfaces. Example: the pressure difference across the airfoil creates lift.

PROP-021

Why do large cinematic drones use larger props?

- A. Higher Efficiency B. Lower Disk Loading C. More Lift **D. All of the Above ✓**

Why: Large cinematic drones use larger props because they are more efficient, give lower disk loading and produce more lift — all of the above. Example: a big camera drone gets long flight time thanks to large, slow-turning props.

PROP-022

What causes prop wash oscillations?

- A. Turbulent Air ✓** B. ESC Firmware C. Battery Brand D. GPS Error

Why: Prop wash oscillations are caused by turbulent air, as the quad flies through its own disturbed air. ESC firmware, battery brand and GPS error are not the cause. Example: wobbles when descending through your own prop wash.

PROP-023

Tiny Whoops usually use:

- A. Small Lightweight Props ✓** B. 15-inch Props C. Helicopter Blades D. Aircraft Wings

Why: Tiny whoops use small, lightweight props that match their small motors and low power. 15-inch props or aircraft wings would be far too large. Example: a whoop has props only about 31mm across.

PROP-024

What is prop wash?

- A. Disturbed Airflow Produced by Propeller ✓ B. Water Cleaning Method C. ESC Cooling D. GPS Drift

Why: Prop wash is the disturbed, turbulent airflow produced by the propellers. It is not water, ESC cooling or GPS drift. Example: flying in your own prop wash shows up as vibration or oscillation in the video.

PROP-025

What is drag?

- A. Forward Force B. Resistance to Motion ✓ C. Lift Force D. Magnetic Force

Why: Drag is the resistance the air puts up against motion. It grows quickly with speed, which is why fast flying uses more battery. It is not a forward, lift or magnetic force. Example: pushing a quad to high speed makes it fight much more drag than a slow cruise, so flight time drops.

PROP-026

What is angle of attack?

- A. Angle Between Airflow and Blade Chord ✓ B. Motor Angle C. Camera Angle D. GPS Angle

Why: Angle of attack is the angle between the oncoming airflow and the blade's chord line. It is not a motor, camera or GPS angle. Example: too high an angle of attack makes the blade stall.

PROP-027

What happens if prop diameter increases?

- A. More Air Moved B. More Thrust C. More Current Draw D. All of the Above ✓

Why: A larger prop diameter moves more air, gives more thrust and needs more current, so all of the listed things happen. Example: going from 5-inch to 6-inch props increases thrust and amps.

PROP-028

Most FPV propellers are made from:

- A. Steel B. Carbon Fiber Reinforced Plastic ✓ C. Copper D. Aluminum

Why: Most FPV props are made from carbon-fibre-reinforced plastic or similar plastics that are light and stiff. Steel, copper and aluminium are too heavy or unsuitable. Example: hobby props are usually glass- or carbon-reinforced plastic.

PROP-029

What is thrust?

- A. Force Moving Drone Forward/Upward ✓ B. Gravity C. Drag D. Weight

Why: Thrust is the force that moves the drone forward or upward, produced by the propellers. Gravity, drag and weight are forces that oppose it. Example: when thrust is greater than weight, the drone climbs.

PROP-030

Bi-blade props have:

- A. 1 Blade **B. 2 Blades ✓** C. 3 Blades D. 4 Blades

Why: Bi-blade props have two blades. With fewer blades there is less drag and less current draw, so they are efficient, but they grip the air less. Example: a 7-inch long-range quad often uses bi-blade props to squeeze out more flight time per battery.

PROP-031

A propeller is essentially a:

- A. Rotating Wing ✓** B. Battery C. Motor D. ESC

Why: A propeller is essentially a rotating wing: each blade is an airfoil that creates lift as it spins. It is not a battery, motor or ESC. Example: it works on the same principle as a plane's wing.

PROP-032

Lower pitch props generally provide:

- A. Better Efficiency B. Lower Current Draw C. Smoother Control **D. All of the Above ✓**

Why: Lower pitch props generally give better efficiency, lower current draw and smoother control — all of the above. Example: long-range builds choose lower pitch props for efficiency.

PROP-033

Why do racers often use high-pitch props?

- A. Faster Acceleration B. Higher Top Speed C. Better Grip **D. All of the Above ✓**

Why: Racers use high-pitch props for faster acceleration, higher top speed and better grip in the air — all of the above (at the cost of more current). Example: a racing build runs aggressive high-pitch props for speed.

PROP-034

What is a tip vortex?

- A. Battery Failure **B. Circular Airflow at Blade Tip ✓** C. ESC Error D. GPS Drift

Why: A tip vortex is circular airflow that forms at the blade tip as high-pressure air leaks to low-pressure air. It is not a battery or ESC problem. Example: tip vortices waste energy and add noise.

PROP-035

Higher pitch props generally provide:

- A. More Top Speed ✓** B. Less Speed C. Less Airflow D. Less Thrust

Why: A higher pitch prop moves further forward per rotation, so it gives more top speed, at the cost of more current and less snappy low-speed response. Example: a 5x5 prop reaches a higher top speed than a 5x4.3 but punches out less quickly and heats the motors more.

PROP-036

Why are propeller blades twisted from root to tip?

- A. Maintain Efficient Angle of Attack ✓** B. Improve Appearance C. Reduce Weight Only D. Increase Hub Size

Why: Blades are twisted so the angle of attack stays efficient along the whole blade: the tip moves faster than the root, so it needs a smaller angle. It is not about appearance or hub size. Example: the blade has a steeper angle near the hub and a flatter one at the tip.

PROP-037

Carbon fiber props are generally:

- A. Stiffer B. Stronger C. More Efficient **D. All of the Above ✓**

Why: Carbon fibre props are stiffer, stronger and more efficient, so all of those apply. A stiff blade keeps its shape under load, so the angle of attack stays correct and thrust stays consistent. Example: carbon props feel crisper than flexible plastic ones, but they can crack on impact.

PROP-038

What is an airfoil?

- A. Propeller Blade Cross Section ✓** B. ESC Component C. Battery Connector D. Receiver Type

Why: An airfoil is the cross-section shape of a propeller blade (or wing) designed to create lift. It is not an ESC part, battery connector or receiver. Example: cut across a blade and you see its airfoil.

PROP-039

Tri-blade props have:

- A. 2 Blades **B. 3 Blades ✓** C. 4 Blades D. 5 Blades

Why: Tri-blade props have three blades. They are the common balance of thrust, grip and efficiency. More blades give more bite and thrust but need more current. Example: most freestyle and racing 5-inch quads fly tri-blade props.

PROP-040

What is the center of a propeller called?

- A. Rotor B. Bell **C. Hub ✓** D. Shaft

Why: The centre of a propeller is called the hub, where it mounts onto the motor. Rotor, bell and shaft are motor parts. Example: the prop hub slides over the motor shaft.

PROP-041

What creates low pressure above the propeller blade?

- A. Faster Airflow ✓** B. Battery Voltage C. ESC Signals D. GPS

Why: Faster airflow over the top of the blade creates low pressure there (Bernoulli). Battery voltage, ESC signals and GPS do not. Example: lower pressure above and higher pressure below pulls the blade upward.

PROP-042

What is the primary purpose of a drone propeller?

- A. Store Energy **B. Generate Thrust ✓** C. Generate GPS Signals D. Control ESC

Why: The main purpose of a drone propeller is to generate thrust. It does not store energy, generate GPS signals or control the ESC. Example: spinning props push air down to lift the drone.

PROP-043

What does a "5-inch propeller" refer to?

- A. Blade Width **B. Prop Diameter ✓** C. Hub Diameter D. Shaft Diameter

Why: A "5-inch propeller" refers to its diameter, the distance across tip to tip (about 127mm). Pitch is a separate number, such as 5x4.3. Example: a 5-inch frame is built for 5-inch props, and a larger prop will not fit between the arms.

PROP-044

Which direction does air move when thrust is produced?

- A. Upward B. Sideways **C. Backward/Downward ✓** D. No Movement

Why: When thrust is produced air is pushed backward/downward, and the reaction pushes the drone the other way. Upward or sideways is wrong. Example: air goes down from a hovering quad.

PROP-045

What is the fundamental reason an FPV propeller generates thrust?

- A. Creates Pressure Difference and Accelerates Air Mass ✓** B. Uses Magnets C. Uses GPS Signals D. Uses ESC Voltage Directly

Why: A propeller generates thrust by creating a pressure difference and accelerating a mass of air. Magnets, GPS and ESC voltage do not create thrust directly. Example: more air moved faster means more thrust.

PROP-046

What law explains lift generation through pressure difference?

- A. Ohm's Law **B. Bernoulli's Principle ✓** C. Kirchhoff's Law D. Hooke's Law

Why: Bernoulli's principle explains lift through the pressure difference caused by different air speeds. Ohm's, Kirchhoff's and Hooke's laws apply to electricity and springs. Example: lower pressure above the blade, higher below.

PROP-047

What is induced drag?

- A. Drag Produced While Generating Lift ✓** B. Battery Resistance C. ESC Heat D. Motor Friction

Why: Induced drag is the drag produced as a by-product of generating lift. It is not battery resistance, ESC heat or motor friction. Example: tip vortices contribute to induced drag.

PROP-048

Which propeller material generally offers the highest stiffness?

- A. Nylon B. Polycarbonate **C. Carbon Fiber Composite ✓** D. Rubber

Why: Carbon fibre composite offers the highest stiffness among these materials. Nylon, polycarbonate and rubber are more flexible. Example: stiff carbon props keep blade shape at high RPM.

PROP-049

Why are propeller tips often swept?

- A. Reduce Noise B. Reduce Tip Vortices C. Improve Efficiency **D. All of the Above ✓**

Why: Swept tips reduce noise, reduce tip vortices and improve efficiency, so all of the above. Curving the tip lets the blade slice the air more cleanly. Example: low-noise props have curved tips, which makes cinematic builds quieter and a little more efficient.

PROP-050

What is the main aerodynamic purpose of an airfoil shape?

- A. Generate Pressure Difference B. Generate Lift C. Increase Efficiency **D. All of the Above ✓**

Why: An airfoil shape creates a pressure difference, generates lift and increases efficiency — all of the above. Example: well-designed airfoils make more thrust for less power.

5. Video Transmitter and Camera Systems

Seeing through the drone: cameras, sensors, VTX power and channels, antennas, polarisation and analog vs digital video.

50 questions

VTXCAM-001

What causes multipath interference?

- A. Signal Reflections ✓** B. Battery Voltage C. ESC Noise D. GPS Error

Why: Multipath interference happens when the video signal bounces off buildings, walls or the ground and arrives at the goggles several times by different paths, so the copies disturb each other. Battery voltage, ESC noise and GPS are not the cause. Example: flying near a metal building makes the picture break up even though you are close.

VTXCAM-002

25mW is commonly used for:

- A. Indoor Flying ✓** B. Satellites C. Long Range D. Cellular Towers

Why: 25mW is a very low transmit power, suited to short-range use such as indoor flying and racing gates. It is far too weak for long range or reaching satellites or towers. Example: flying around a gym or warehouse with a 25mW VTX is plenty.

VTXCAM-003

Why are different VTX channels used?

- A. Avoid Interference ✓** B. Increase Battery Life C. Improve GPS D. Improve Motors

Why: Different VTX channels let pilots fly at the same time on separate frequencies, which avoids interference. Channels do not extend battery life or improve GPS or motors. Example: four pilots racing together each pick a different channel.

VTXCAM-004

Which component converts light into electronic signals?

- A. Camera Sensor ✓** B. VTX C. Antenna D. ESC

Why: The camera sensor converts light into electronic signals. The VTX transmits video, the antenna radiates it and the ESC drives motors. Example: the CMOS chip behind the lens is what actually "sees".

VTXCAM-005

What is bitrate in digital FPV systems?

- A. Amount of Video Data per Second ✓** B. Battery Voltage C. VTX Frequency D. Camera Weight

Why: Bitrate is the amount of video data sent per second in a digital FPV system. It is not battery voltage, frequency or camera weight. Example: a higher bitrate such as 50Mbps carries more picture detail than 25Mbps.

VTXCAM-006

What happens if RHCP and LHCP are mixed?

- A. Signal Loss ✓** B. More Range C. More Power D. Better Quality

Why: A right-hand circular (RHCP) antenna and a left-hand circular (LHCP) antenna are opposite polarisations, so mixing them causes a big signal loss. It does not add range, power or quality. Example: an LHCP VTX antenna with RHCP goggles gives a weak, noisy picture.

VTXCAM-007

What primarily limits FPV transmission range?

- A. Signal-to-Noise Ratio ✓** B. ESC Current C. Motor KV D. Propeller Pitch

Why: FPV range is mainly limited by the signal-to-noise ratio — how strong the signal is compared with background noise. ESC current, motor KV and prop pitch have no effect. Example: as you fly away, the signal weakens until noise overwhelms it and the picture breaks up.

VTXCAM-008

The Friis Transmission Equation is primarily related to:

- A. RF Signal Propagation ✓** B. Motor Torque C. ESC Efficiency D. Battery Charging

Why: The Friis transmission equation describes how radio signals weaken as they travel, i.e. RF signal propagation. It is not about motors, ESCs or charging. Example: it explains why doubling the distance cuts received power by roughly a quarter.

VTXCAM-009

Why do long-range pilots use higher VTX power?

- A. Extend Signal Distance ✓** B. Improve Battery Life C. Reduce Heat D. Increase RPM

Why: Long-range pilots increase VTX power to extend how far the video signal reaches. It does not save battery, lower heat or raise RPM (it actually uses more power). Example: switching from 25mW to 800mW can help keep video at distance.

VTXCAM-010

Omnidirectional antennas are preferred when:

- A. Flying in Multiple Directions ✓** B. Flying Only Straight C. Flying Indoors Only D. Battery Charging

Why: Omnidirectional antennas receive and send in all directions, so they suit flying in many directions. They do not suit charging or flying only indoors. Example: freestyle pilots move around the sky and use omni antennas to keep the signal.

VTXCAM-011

Circular Polarized antennas are used because:

- A. Reduce Multipath Interference ✓** B. Increase Voltage C. Reduce Current D. Improve GPS

Why: Circularly polarised antennas reduce multipath interference, because reflected signals reverse their polarisation and are rejected. They do not raise voltage, cut current or improve GPS. Example: CP antennas give a cleaner picture around buildings.

VTXCAM-012

What happens if two pilots use the same VTX frequency?

- A. Better Signal **B. Video Interference ✓** C. More Range D. Faster Drone

Why: Two pilots on the same VTX frequency transmit over each other, causing video interference. It does not improve signal, range or speed. Example: when a friend turns on at the same channel, both pictures turn to static.

VTXCAM-013

A camera lens is used to:

- A. Focus Light onto Sensor ✓** B. Increase Voltage C. Generate RF D. Measure GPS

Why: A camera lens focuses light onto the sensor. It decides how wide the view is and how sharp the picture looks. It does not raise voltage, generate RF or measure GPS. Example: a wide-angle lens shows more of the scene, while a dirty or cracked lens makes the whole video blurry.

VTXCAM-014

What happens if an obstacle blocks the Fresnel Zone?

- A. Signal Quality Decreases ✓** B. Signal Improves C. Video Resolution Doubles D. Motor Speed Increases

Why: If an obstacle blocks the Fresnel zone (the region around the direct line between antennas), signal quality decreases. It does not improve signal or video. Example: flying low behind trees or a hill drops the video quality even without a fully blocked line of sight.

VTXCAM-015

Why does video compression introduce latency?

- A. Encoding and Decoding Takes Time ✓** B. More Antennas C. More Motors D. More ESCs

Why: Compression introduces latency because encoding the video at the drone and decoding it in the goggles takes time. More antennas, motors or ESCs are unrelated. Example: digital systems have slightly more delay than analog because of this processing.

VTXCAM-016

What is image latency?

- A. Delay Between Reality and Display ✓** B. Video Resolution C. Battery Delay D. GPS Delay

Why: Image latency is the delay between something happening in the real world and you seeing it on your display. It is not resolution, battery delay or GPS delay. Example: lower latency makes the quad feel more connected.

VTXCAM-017

Why are patch antennas used for long-range FPV?

- A. Higher Directional Gain ✓** B. Lower Weight C. Lower Cost D. More Current

Why: Patch antennas focus the signal in one direction, giving higher directional gain and so more range that way. They are not for weight, cost or current. Example: pilots use a patch on their goggles pointed at the drone for distance.

VTXCAM-018

Which FPV system generally provides the highest image quality?

- A. Analog **B. DJI O4 ✓** C. 5.8GHz Analog D. AIO Camera

Why: Among these options, DJI O4 generally gives the highest digital image quality; analog systems have lower quality. Example: digital HD video shows far more detail than analog.

VTXCAM-019

What is a VTX channel?

- A. Specific Frequency Slot ✓** B. ESC Output C. Camera Mode D. Motor Speed

Why: A VTX channel is a specific frequency slot used for transmission. It is not an ESC output, camera mode or motor speed. Example: Raceband channel R1 is one slot such as 5658MHz.

VTXCAM-020

Which statement best explains FPV video transmission?

- A. Camera Captures Light → Encodes Video → VTX Modulates RF → Antenna Radiates Signal → Goggles Decode Video ✓** B. Battery Directly Sends Video C. ESC Creates Video D. Motors Generate Video

Why: Video path: the camera captures light, encodes video, the VTX modulates it onto RF, the antenna radiates it, and the goggles decode and display it. The battery, ESC or motors do not create video. Example: each step in that chain must work for you to see a picture.

VTXCAM-021

What is signal attenuation?

- A. Reduction of Signal Strength Over Distance ✓** B. Increased Signal Power C. Better Video Quality D. Battery Drain

Why: Signal attenuation is the reduction in signal strength as it travels. It is not an increase in power or a battery issue. Example: attenuation is why the picture degrades as you fly further away.

VTXCAM-022

What is the Fresnel Zone?

- A. RF Propagation Region Around Direct Signal Path ✓** B. Camera Lens Type C. ESC Component D. Battery Design

Why: The Fresnel zone is the region of space around the direct line between transmitter and receiver that radio waves travel through. It is not a lens, ESC part or battery design. Example: keeping this zone clear of obstacles keeps the signal strong.

VTXCAM-023

What is the purpose of image sensors such as CMOS?

- A. Convert Photons into Electrical Signals ✓** B. Generate RF C. Power Motors D. Store Battery Energy

Why: An image sensor such as a CMOS chip converts photons (light) into electrical signals. It does not generate RF or power motors. Example: light hitting the CMOS sensor becomes the video signal.

VTXCAM-024**Most FPV VTX systems operate on:**

- A. 433MHz B. 2.4GHz **C. 5.8GHz ✓** D. 24GHz

Why: Most FPV VTX systems operate on 5.8GHz. Control links use other bands such as 2.4GHz and 900MHz. Example: your goggles are tuned to a 5.8GHz channel to receive video.

VTXCAM-025**What is the primary purpose of an FPV camera?**

- A. Generate GPS **B. Capture Real-Time Video ✓** C. Control ESC D. Charge Battery

Why: An FPV camera captures real-time video of what the drone sees. It does not generate GPS, control ESCs or charge batteries. Example: the camera feed is what you see in the goggles.

VTXCAM-026**What component radiates the video signal?**

- A. ESC **B. Antenna ✓** C. Battery D. Receiver

Why: The antenna radiates the video signal as radio waves. The VTX makes the signal, but the antenna sends it out; the ESC, battery and receiver do not. Example: a broken antenna sharply cuts your video range, and you must always fit the antenna before powering the VTX.

VTXCAM-027**What is video resolution?**

- A. Image Detail Level ✓** B. Battery Capacity C. VTX Voltage D. ESC Speed

Why: Video resolution is how much detail an image has, measured in pixels. It is not battery capacity, VTX voltage or ESC speed. Example: 1080p shows more detail than 480p.

VTXCAM-028**What is dynamic range?**

- A. Ability to Handle Bright and Dark Areas ✓** B. Maximum VTX Power C. Maximum Flight Time D. Motor Torque

Why: Dynamic range is a camera's ability to handle bright and dark areas in the same scene. It is not VTX power, flight time or motor torque. Example: good dynamic range lets you see both the bright sky and a dark tree line.

VTXCAM-029**What is the fundamental purpose of an FPV VTX and Camera system?**

- A. Convert Real-World Visual Information into Wireless Video Data for the Pilot ✓** B. Generate Thrust C. Control Motors D. Navigate GPS

Why: The VTX and camera system converts real-world visual information into wireless video data the pilot can see. It does not generate thrust, control motors or navigate. Example: it lets you see the world from the drone's viewpoint.

VTXCAM-030**Modern digital FPV systems use:**

- A. Video Compression ✓** B. Morse Code C. PWM Signals D. Analog Filters Only

Why: Modern digital FPV systems use video compression to fit HD video into the radio bandwidth. They do not use Morse code or analog filters alone. Example: compression is why digital systems can send HD video over a limited link.

VTXCAM-031**RHCP stands for:**

- A. Right Hand Circular Polarization ✓** B. Radio High Current Power C. Rotor High Circular Phase D. Receiver Hardware Control Protocol

Why: RHCP stands for Right Hand Circular Polarisation: the radio wave rotates clockwise as it travels. VTX and goggles antennas should use the same polarisation. Example: most FPV antennas are RHCP, so an RHCP VTX antenna should be paired with an RHCP goggle antenna.

VTXCAM-032**What does FPS mean?**

- A. Frames Per Second ✓** B. Flight Power System C. Frequency Per Signal D. Frame Processing Speed

Why: FPS means Frames Per Second, how many video frames are shown each second. Higher FPS looks smoother and makes fast movement easier to follow. Example: 60fps video looks smoother than 30fps, which matters when flying fast near obstacles.

VTXCAM-033**What does VTX stand for?**

- A. Video Transmission System **B. Video Transmitter ✓** C. Visual Transfer Unit D. Voltage Transmitter

Why: VTX stands for Video Transmitter. It takes the picture from the camera and broadcasts it on 5.8GHz to your goggles. Example: it is the small board beside the antenna, and its output power can be set from about 25mW up to 1000mW.

VTXCAM-034**What is the purpose of FPV goggles?**

- A. Receive Video Feed ✓** B. Power Motors C. Control Drone D. Store Data

Why: FPV goggles receive the video feed and show it to the pilot. They do not power motors, control the drone or store data primarily. Example: you see the drone's view in the goggles.

VTXCAM-035**LHCP stands for:**

- A. Left Hand Circular Polarization ✓** B. Low Heat Camera Processing C. Long Horizon Camera Position D. Local High Current Power

Why: LHCP stands for Left Hand Circular Polarisation: the wave rotates anticlockwise, the opposite of RHCP. Mixing the two loses a lot of signal. Example: some pilots use LHCP on purpose so their video does not clash with others who fly RHCP.

VTXCAM-036**Why does increasing VTX power not double range?**

- A. RF Propagation is Nonlinear ✓** B. Battery Limits C. Camera Limits D. GPS Limits

Why: Doubling VTX power does not double range because radio propagation is nonlinear — signal strength falls rapidly with distance. Battery, camera or GPS limits are not the reason. Example: going from 200mW to 400mW gives only a small range gain.

VTXCAM-037**Which has higher resolution?**

- A. Analog **B. DJI O4 ✓** C. NTSC D. PAL

Why: DJI O4 digital video has higher resolution than analog, NTSC or PAL, which are analog standards with low resolution. Example: digital HD shows much more detail than analog.

VTXCAM-038**What does RF stand for?**

- A. Radio Frequency ✓** B. Remote Function C. Rotor Force D. Rotational Frequency

Why: RF stands for Radio Frequency, the radio waves used to carry signals through the air. Example: your 2.4GHz control link and your 5.8GHz video link are both RF links, working on different frequency bands.

VTXCAM-039**What is polarization?**

- A. Antenna Signal Orientation ✓** B. Battery Type C. Camera Setting D. ESC Mode

Why: Polarisation is the orientation of an antenna's signal (for example linear or circular). It is not a battery type, camera setting or ESC mode. Example: matching polarisation between VTX and goggles antennas gives the best signal.

VTXCAM-040**Which FPV component captures light?**

- A. Camera Sensor ✓** B. VTX C. Antenna D. ESC

Why: The camera sensor (often a CMOS chip) captures the light and converts it into electrical signals. The VTX, antenna and ESC do not capture light. Example: a larger sensor collects more light, so the picture is cleaner in dim conditions.

VTXCAM-041**Which FPV system is famous for ultra-low latency?**

- A. Analog **B. HDZero ✓** C. DJI O4 D. Walksnail

Why: HDZero is known for ultra-low latency: it is a digital system designed to keep the delay between reality and your goggles as small as analog. Most digital systems add more delay. Example: racers like HDZero because they see the track almost instantly.

VTXCAM-042**What happens if a VTX runs without an antenna?**

- A. Nothing B. Better Signal **C. Possible Overheating/Damage ✓** D. Faster Video

Why: Running a VTX without an antenna can overheat or damage it, as the power has nowhere to radiate. It does not improve signal. Example: never power up a VTX on the bench without an antenna attached.

VTXCAM-043**Which FPV system is analog?**

- A. DJI O4 B. Walksnail C. HDZero **D. Analog FPV ✓**

Why: Analog FPV is the analog system: it sends analog video (NTSC or PAL) with very low delay but low resolution and visible static when the signal weakens. DJI O4, Walksnail and HDZero are digital. Example: classic analog is cheap and reacts instantly, but looks grainy.

VTXCAM-044**What is the primary purpose of a VTX?**

- A. Control Motors **B. Send Video to Goggles ✓** C. Power ESC D. Receive GPS

Why: The VTX's primary purpose is to send video to the goggles. It does not control motors, power the ESC or receive GPS. Example: it transmits the camera's picture over 5.8GHz.

VTXCAM-045**Why do FPV cameras perform better at night when sensor size increases?**

- A. Collect More Light ✓** B. Increase Voltage C. Improve GPS D. Reduce Weight

Why: A larger sensor collects more light, improving night performance. It does not raise voltage, improve GPS or reduce weight. Example: bigger sensors give a cleaner picture in dim light.

VTXCAM-046**Higher bitrate generally provides:**

- A. Better Image Quality ✓** B. Lower Resolution C. Less Data D. Less Detail

Why: A higher bitrate generally gives better image quality because more video data is sent. It does not lower resolution or detail. Example: increasing the bitrate gives a sharper image.

VTXCAM-047**What is measured in mW on a VTX?**

- A. Voltage **B. Power Output ✓** C. Frequency D. Current

Why: mW (milliwatts) on a VTX is its power output. Voltage, frequency and current use other units. More power can extend range but also makes more heat and may be limited by local rules. Example: 25mW is enough indoors, while 600mW or more is used for distance.

VTXCAM-048

The video signal path is:

- A. Camera → VTX → Antenna → Goggles ✓** B. Battery → ESC → Camera C. GPS → Receiver → FC D. Motor → ESC → Goggles

Why: The video signal path is camera, then VTX, then antenna, then goggles. The other paths describe power, GPS or motor routes. Example: the camera feeds the VTX which radiates to your goggles.

VTXCAM-049

Which FPV system generally has the longest range?

- A. Analog 25mW **B. DJI O4 ✓** C. Tiny Whoop Camera D. AIO Camera

Why: Among these, DJI O4 generally has the longest range, since it is a digital system with strong error correction. A 25mW analog link or tiny whoop camera has much less range. Example: digital systems keep a usable picture further out.

VTXCAM-050

What does FOV mean?

- A. Field of View ✓** B. Flight Output Voltage C. Frame of Vision D. Frequency Output Value

Why: FOV stands for Field of View, the angle of the scene the camera captures. A wide FOV shows more of the surroundings and gives a stronger sense of speed; a narrow FOV looks more zoomed in. Example: freestyle pilots like wide FOV for awareness, while cinematic pilots often choose a narrower one.

6. FPV Assembly and Soldering

Building reliably: soldering technique, wiring, testing before power-up, frames and professional build habits.

50 questions

ASSEMBLY-001

Which soldering mistake most commonly damages FC pads?

- A. Using flux **B. Repeated reheating and pulling wires ✓** C. Using silicone wires D. Cleaning with IPA

Why: Pads on a flight controller are thin copper bonded to the board. Heating the same pad again and again, or pulling on a wire while it is hot, lifts the pad off the PCB. Flux, silicone wire and cleaning with IPA are normal good practice. Example: tugging a wire off a pad that is still molten rips the pad away and ruins the board.

ASSEMBLY-002

Why are battery leads generally kept short?

- A. Lower weight only **B. Reduce voltage spikes and resistance ✓** C. Improve GPS D. Better aesthetics

Why: Short battery leads have less resistance and inductance, so they cause smaller voltage spikes. Weight, GPS and looks are not the main reasons. Example: long leads on a powerful quad can let spikes reach the ESC and damage it, which is why a capacitor is also added.

ASSEMBLY-003

What causes most beginner build failures?

- A. Expensive components **B. Poor workmanship ✓** C. Frame weight D. Propeller color

Why: Most beginner build failures come from poor workmanship, such as bad solder joints, short circuits or loose screws. Expensive parts, frame weight and propeller colour rarely cause failure. Example: a single cold joint on a battery lead can cause a mid-air power loss.

ASSEMBLY-004

A student increases iron temperature from 350°C to 480°C to solder faster. What is the biggest risk?

- A. Better conductivity **B. Pad delamination and PCB damage ✓** C. Reduced oxidation D. Stronger solder joints

Why: Using too high an iron temperature (480°C instead of about 350°C) can lift pads and delaminate the PCB, damaging the board. It does not give better conductivity or stronger joints. Example: a very hot iron held on a pad for a second too long lifts the copper and ruins the board.

ASSEMBLY-005

Which tool is most useful for checking short circuits before connecting a battery?

- A. GPS **B. Multimeter ✓** C. Oscilloscope D. Receiver

Why: A multimeter in continuity mode is the most useful tool for checking for short circuits before connecting a battery. A GPS, oscilloscope or receiver does not do this. Example: touching the probes to the battery pads and hearing a beep means a dangerous short.

ASSEMBLY-006

What should determine ESC current rating selection?

- A. Frame color **B. Expected motor current draw ✓** C. GPS size D. Antenna gain

Why: Choose the ESC current rating based on the motors' expected current draw. Frame colour, GPS size and antenna gain are irrelevant. Example: if the motors may draw 35A each at full throttle, you need an ESC rated above that.

ASSEMBLY-007

Why should antennas always be installed before powering a VTX?

- A. Better image quality **B. Prevent RF damage from reflected power ✓** C. Increase resolution D. Increase FPS

Why: Always attach the antenna before powering a VTX, otherwise the transmitted power reflects back into the VTX and can overheat and damage it. It does not improve image quality. Example: powering a VTX on the bench without an antenna can burn out its output stage in seconds.

ASSEMBLY-008

Why is flux often more important than increasing soldering temperature?

- A. It reduces wire resistance **B. It removes oxidation and improves wetting ✓** C. It increases current carrying capacity D. It cools the solder

Why: Flux removes oxidation and helps solder flow and wet the surface, which gives a good joint without needing extra heat. It does not reduce wire resistance or cool the solder. Example: adding flux lets solder flow onto a pad at normal temperature instead of balling up.

ASSEMBLY-009

A solder joint looks dull gray instead of shiny. What should be suspected first?

- A. Strong joint **B. Cold joint ✓** C. Excessive copper D. Excessive current capacity

Why: A dull grey joint is the classic sign of a cold solder joint, where the solder did not melt properly or moved while cooling. A good joint is shiny. Example: a grey, crumbly-looking joint can crack and lose connection in flight.

ASSEMBLY-010

The best solder joint is:

- A. Largest possible B. Smallest possible **C. Mechanically secure and electrically sound ✓** D. Most shiny

Why: The best solder joint is mechanically secure and electrically sound — it holds firmly and conducts well. Biggest, smallest or just shiniest is not the goal. Example: a neat joint that does not move when tugged lightly and conducts reliably.

ASSEMBLY-011

A professional builder finishes soldering. What is done before battery connection?

- A. Fly immediately **B. Continuity test ✓** C. Increase VTX power D. Install props

Why: After soldering, a professional does a continuity test before connecting a battery, to catch shorts. Flying immediately, raising VTX power or fitting props skips safety. Example: test across the battery pads for a short before the first power-up.

ASSEMBLY-012

Why do professional builders route wires carefully?

- A. Appearance only **B. Reliability, maintenance, noise reduction ✓** C. Increase battery capacity D. Improve GPS accuracy

Why: Careful wire routing improves reliability, makes maintenance easier and reduces electrical noise. It is not only about looks, and it does not increase battery capacity or GPS accuracy. Example: video wires kept away from power wires give a cleaner picture.

ASSEMBLY-013

A 15-inch heavy-lift drone requires:

- A. Tiny Whoop Electronics **B. Components sized for expected current loads ✓** C. Racing ESCs only D. Smaller motors

Why: A 15-inch heavy-lift drone needs components sized for its expected current loads — high-current ESCs and large motors. Tiny whoop electronics or smaller motors would fail. Example: such a drone may need 80A or larger ESCs and thick battery leads.

ASSEMBLY-014

Why do professional builders use helping hands or PCB holders?

- A. Reduce battery consumption **B. Prevent movement during soldering ✓** C. Increase current flow D. Reduce weight

Why: Helping hands or PCB holders keep parts from moving while you solder, so you can make clean joints. They do not reduce battery use, increase current or cut weight. Example: a steady PCB lets you use both hands for iron and solder.

ASSEMBLY-015

What is the biggest danger of routing motor wires near gyro chips?

- A. Increased lift **B. Electrical noise affecting sensors ✓** C. Better filtering D. Better responsiveness

Why: Motor wires carry fast-switching, high-current pulses that radiate electrical noise. Running them near the gyro can disturb its readings. They do not add lift or improve filtering. Example: noisy gyro data can cause twitchy flight or oscillations.

ASSEMBLY-016

Which connection is most critical in a digital FPV system?

- A. Camera signal path ✓** B. LED strip C. GPS D. Buzzer

Why: In a digital FPV system the camera signal path is the most critical connection, as any fault stops the video. LED strips, GPS and buzzers are non-essential. Example: a loose camera wire leaves you with no video at all.

ASSEMBLY-017**Why should motor screws be checked carefully?**

- A. Aesthetic reasons **B. Too-long screws can touch windings ✓** C. ESC protocol issues D. GPS interference

Why: Screws that are too long can poke through the motor base and touch the windings, shorting them or damaging the wire insulation. It is not about looks, ESC protocol or GPS. Example: use the screw length recommended for the motor.

ASSEMBLY-018**What should be selected first during a build?**

- A. Stickers **B. Flight goals and requirements ✓** C. Battery strap D. Receiver color

Why: Start a build by deciding the flight goals and requirements (freestyle, racing, long range) because they drive every part choice. Stickers, straps and receiver colour come last. Example: a long-range goal leads to a 7-inch frame and efficient motors.

ASSEMBLY-019**A VTX powers up but no video appears. First suspect?**

- A. Motor direction **B. Camera signal path ✓** C. GPS baud rate D. ESC timing

Why: If the VTX powers up but there is no video, suspect the camera signal path first (wiring or camera). Motor direction, GPS baud rate and ESC timing do not affect video. Example: check the camera's signal wire and connector.

ASSEMBLY-020**When soldering XT60 connectors, the greatest danger is:**

- A. High resistance **B. Connector melting from prolonged heating ✓** C. Too much flux D. Wire flexibility

Why: XT60 connectors have plastic bodies that melt if heated too long, so the biggest danger is melting the connector. Resistance, flux and wire flexibility are smaller concerns. Example: use a hot iron briefly and hold the connector steady.

ASSEMBLY-021**A digital VTX suddenly shuts down in flight. Most likely?**

- A. Low battery **B. Overheating ✓** C. High GPS count D. Motor KV

Why: A digital VTX that suddenly shuts down in flight is most likely overheating, since it runs hot without airflow. Low battery, GPS count and motor KV are not the usual cause. Example: running a digital VTX on the ground for long without airflow can trigger thermal shutdown.

ASSEMBLY-022**A drone powers through USB but not battery. First suspect?**

- A. GPS **B. Battery power path issue ✓** C. Camera D. Receiver

Why: If a drone powers over USB but not from the battery, suspect the battery power path (leads, XT60 joints, power pads). GPS, camera and receiver are not the first suspect. Example: a cold joint on a battery lead explains it.

ASSEMBLY-023

What is the purpose of strain relief?

- A. Improve aesthetics **B. Prevent wire fatigue and breakage ✓** C. Improve VTX range D. Improve ESC efficiency

Why: Strain relief protects wires from fatigue and breaking where they bend or are tugged. It does not improve looks, VTX range or ESC efficiency. Example: a strain-relieved wire does not snap off at the pad after repeated flexing.

ASSEMBLY-024

A builder owns only one test instrument. Which should it be?

- A. Spectrum Analyzer **B. Multimeter ✓** C. GPS D. RF Meter

Why: If you own only one test instrument it should be a multimeter, which checks continuity, voltage and resistance. A spectrum analyser or RF meter is specialised. Example: it finds shorts before the first power-up.

ASSEMBLY-025

Which frame typically survives crashes better?

- A. Integrated-arm frame **B. Replaceable-arm frame ✓** C. Plastic frame D. Thin frame

Why: A frame with replaceable arms survives crashes better, because a broken arm can be swapped instead of replacing the frame. Integrated-arm, plastic or thin frames break more easily. Example: swap a cracked arm in minutes.

ASSEMBLY-026

What is the safest way to remove excess solder from a pad?

- A. Knife B. Wire cutter **C. Solder wick ✓** D. Screwdriver

Why: Solder wick (desoldering braid) is the safest way to remove excess solder from a pad. A knife, cutters or screwdriver can damage the pad. Example: lay the wick on the pad, heat it and the solder soaks up.

ASSEMBLY-027

Which is generally the most reliable soldering sequence?

- A. Heat solder → Heat pad → Add wire B. Heat wire only **C. Heat pad and wire together, then apply solder ✓**
D. Melt solder first and drop it onto the pad

Why: The most reliable sequence is to heat the pad and wire together and then apply solder, so it flows into the joint properly. Melting solder first and dropping it on makes a cold joint. Example: touch iron to pad and wire, feed solder to the joint.

ASSEMBLY-028

Which frame characteristic most affects resonance?

- A. Carbon design and stiffness ✓** B. VTX power C. GPS brand D. Camera angle

Why: Frame resonance mostly depends on the carbon design and stiffness. VTX power, GPS brand and camera angle do not matter. Example: a flexible frame can resonate and cause vibrations in the video.

ASSEMBLY-029

A professional build is characterized by:

- A. Expensive parts **B. Reliability and serviceability ✓** C. Highest KV motors D. Largest battery

Why: A professional build is characterised by reliability and serviceability (easy to repair). Expensive parts, the highest KV or the biggest battery do not make it professional. Example: neat wiring that is easy to maintain.

ASSEMBLY-030

A motor stutters on arm. Most likely cause?

- A. Broken GPS **B. Damaged motor wire or ESC phase issue ✓** C. VTX overheating D. Camera lens issue

Why: A motor that stutters on arming most likely has a damaged motor wire or an ESC phase problem. GPS, VTX overheating or camera lens issues do not affect motor start-up. Example: check solder joints and wires on that motor.

ASSEMBLY-031

Why should video wires be kept away from power wires when possible?

- A. Weight reduction **B. Noise reduction ✓** C. Better airflow D. Better soldering

Why: Keeping video wires away from power wires reduces electrical noise in the picture. It is not about weight, airflow or soldering. Example: noise lines in the video often come from wires running too close together.

ASSEMBLY-032

What should be verified before installing propellers?

- A. Motor direction B. Receiver operation C. Failsafe **D. All of the Above ✓**

Why: Before installing propellers verify motor direction, receiver operation and failsafe — so "all of the above". Example: with props off, check each motor spins the right way and the quad disarms on signal loss.

ASSEMBLY-033

What is the purpose of heat shrink tubing?

- A. Reduce resistance **B. Insulate exposed connections ✓** C. Increase current capacity D. Cool wires

Why: Heat shrink tubing insulates exposed connections. It does not reduce resistance, increase current capacity or cool wires. Example: it covers a solder joint so it cannot short.

ASSEMBLY-034

What is the primary purpose of a low-ESR capacitor on the ESC?

- A. Increase RPM **B. Reduce voltage spikes and noise ✓** C. Improve GPS lock D. Increase VTX power

Why: A low-ESR capacitor on the ESC reduces voltage spikes and noise. It does not increase RPM, improve GPS lock or raise VTX power. Example: it protects against spikes that can damage the ESC or cause video noise.

ASSEMBLY-035

A 7-inch long-range frame generally prioritizes:

- A. Efficiency and stability ✓** B. Maximum agility C. Tiny size D. Indoor flying

Why: A 7-inch long-range frame generally prioritises efficiency and stability, not maximum agility or tiny size. Example: it is built for long flight times and smooth flying.

ASSEMBLY-036

Which tool provides the biggest improvement in soldering quality?

- A. Expensive drone frame **B. Quality temperature-controlled soldering station ✓** C. Larger battery D. Better propellers

Why: A quality temperature-controlled soldering station gives the biggest improvement in soldering quality. A better frame, larger battery or propellers do not. Example: it holds a steady temperature for clean joints.

ASSEMBLY-037

Why are deadcat frames popular for cinematic builds?

- A. Faster motors **B. Props stay out of camera view ✓** C. Stronger VTX signal D. Better GPS

Why: Deadcat frames are popular for cinematic builds because the props stay out of the camera's view. They are not about faster motors, VTX signal or GPS. Example: the arms angle back so the props do not appear in the video.

ASSEMBLY-038

A 6S build occasionally resets during punch-outs. What is most likely?

- A. Too much GPS signal **B. Voltage sag/noise issue ✓** C. Excessive VTX power D. Antenna mismatch

Why: A 6S build that resets during punch-outs most likely has a voltage sag or noise issue, where the battery voltage drops briefly under load. GPS signal, VTX power and antenna mismatch do not cause this. Example: a worn battery or missing capacitor.

ASSEMBLY-039

Which build requires greatest attention to vibration isolation?

- A. Cinematic build ✓** B. Racing build C. Tiny Whoop D. Bench test

Why: A cinematic build needs the most careful vibration isolation, because vibration shows up as jello in the video. Racing builds, tiny whoops and bench tests care less. Example: soft-mounted FC and camera for smooth footage.

ASSEMBLY-040

After assembly, continuity exists between XT60 positive and negative pads. What should be done first?

- A. Connect battery B. Power through USB **C. Investigate before applying power ✓** D. Increase capacitor size

Why: If continuity exists between XT60 positive and negative pads, there is a short circuit, so investigate before applying power. Connecting the battery or powering through USB risks damage. Example: look for a solder bridge.

ASSEMBLY-041

A builder repeatedly crashes and breaks antennas. What should change?

- A. VTX power **B. Antenna mounting strategy ✓** C. ESC firmware D. Camera settings

Why: If you keep breaking antennas, change your antenna mounting strategy so they are better protected. VTX power, ESC firmware and camera settings do not prevent breakage. Example: mount the antenna in a protective tube.

ASSEMBLY-042

A beginner wants durability above all else. Which frame characteristic matters most?

- A. Color **B. Carbon thickness and arm strength ✓** C. Sticker quality D. Battery strap color

Why: For durability the most important frame characteristics are carbon thickness and arm strength. Colour and stickers do not matter. Example: thicker arms survive crashes better.

ASSEMBLY-043

A solder joint appears shiny but breaks when the wire is lightly moved. What is the most likely cause?

- A. Excessive flux usage **B. Cold solder joint hidden under surface solder ✓** C. Excessive copper thickness D. High quality solder

Why: A shiny joint that breaks when moved is likely a cold joint hidden under surface solder: solder sits on top without bonding to the pad or wire. Excess flux, copper thickness and quality solder are not the cause. Example: reflow the joint with flux.

ASSEMBLY-044

Which combination is most likely to cause overheating?

- A. Small ESC + Extremely Powerful Motors ✓** B. Balanced Components C. Large ESC + Small Motor D. Properly Sized Stack

Why: A small ESC with extremely powerful motors is most likely to overheat because the motors draw more current than the ESC is rated for. Balanced or properly sized stacks do not overheat. Example: a 20A ESC with motors drawing 40A will overheat.

ASSEMBLY-045

What is the primary reason builders pre-tin wires?

- A. Improve appearance B. Reduce solder consumption **C. Improve final bond quality and heat transfer ✓** D. Reduce wire flexibility

Why: Pre-tinning wires improves the final bond quality and heat transfer, so soldering is quicker and cleaner. It is not just about looks, solder use or flexibility. Example: a pre-tinned wire melts into the pad quickly.

ASSEMBLY-046

Which signal is commonly used between FC and ESC today?

- A. VGA B. HDMI **C. DShot ✓** D. Analog Video

Why: DShot is the digital protocol the flight controller uses to tell each ESC how fast to spin its motor. It sends the throttle value as a digital number with error checking, so it is not thrown off by electrical noise or drift like the old analog pulse signals, and it needs no throttle calibration. Example: in the Betaflight configurator you simply choose DShot300 or DShot600 for your motors.

ASSEMBLY-047

Why is a smoke stopper used during first power-up?

- A. Increase voltage **B. Limit catastrophic current during shorts ✓** C. Improve ESC performance D. Improve signal quality

Why: A smoke stopper limits the current during a short on first power-up, preventing catastrophic damage. It does not increase voltage or improve ESC performance. Example: it lights a bulb if there is a short, protecting components.

ASSEMBLY-048

A solder joint forms a perfect sphere sitting on the pad. What does this usually indicate?

- A. Excellent wetting B. Excessive heat transfer **C. Poor wetting or contamination ✓** D. High current capability

Why: A solder joint that forms a perfect sphere on the pad indicates poor wetting or contamination: the solder did not bond to the pad. It is not excellent wetting. Example: clean the pad and add flux.

ASSEMBLY-049

Three motor wires are connected to an ESC. Two wires are swapped. Result?

- A. ESC burns **B. Motor reverses direction ✓** C. Motor stops permanently D. FC fails

Why: Swapping two of the three motor wires reverses the motor's direction. It does not burn the ESC or stop the motor permanently. Example: swap any two wires to correct a motor spinning the wrong way.

ASSEMBLY-050

The mindset of a professional FPV builder is:

- A. Build Fast B. Build Cheap **C. Build Once, Test Thoroughly, Fly Reliably ✓** D. Build Randomly

Why: The professional mindset is to build once, test thoroughly and fly reliably. Building fast, cheap or randomly leads to failures. Example: test every connection before flying.

7. FPV Professional Pilot Certification

Flying skills: racing lines, smooth control, cinematic shots, tracking, anticipation and good habits.

120 questions

PILOT-001

What is a Dolly Shot?

A. Constant Speed Forward Motion ✓ B. Hover C. Roll Only D. Yaw Only

Why: A dolly shot is a cinematic move where the drone travels forward at a constant speed, like a camera on a dolly, so the subject glides smoothly toward or away from the lens. Hovering, roll-only or yaw-only are different moves. Example: flying steadily forward down a row of trees with a smooth, even speed.

PILOT-002

What causes altitude hunting?

A. Excessive throttle corrections ✓ B. GPS C. Camera D. ESC

Why: Altitude hunting is when the drone keeps climbing and sinking because the pilot makes too many throttle corrections. GPS, the camera and the ESC are not the cause. Example: a pilot who keeps adding and cutting throttle never settles into a steady height.

PILOT-003

When making a smooth turn, which controls are combined?

A. Roll + Pitch ✓ B. Throttle Only C. Yaw Only D. Roll Only

Why: A smooth turn combines roll and pitch (bank into the turn while holding forward speed), so those controls work together. Throttle-only, yaw-only or roll-only turns are sloppy. Example: roll right while keeping a little forward pitch to carve a circle.

PILOT-004

The hallmark of expertise is:

A. Consistency ✓ B. Luck C. Hardware D. Power

Why: The hallmark of expertise is consistency: doing the same thing correctly again and again. Luck, hardware and raw power do not make an expert. Example: a pro repeats the same clean lap every time.

PILOT-005

What is race consistency?

A. Repeating Same Lap ✓ B. Maximum Speed C. Biggest Battery D. Highest Altitude

Why: Race consistency means repeating the same lap cleanly lap after lap, not just one fast lap. It is not top speed, battery size or altitude. Example: ten laps within a few hundredths of a second of each other.

PILOT-006

Professional racers brake:

- A. Before Turn ✓** B. During Turn C. After Turn D. Never

Why: Professional racers brake before the turn, so they enter at the right speed and can accelerate out. Braking during or after the turn costs time or causes crashes. Example: slow down on the straight, then turn and power out of the corner.

PILOT-007

The fastest path through a corner is called:

- A. Racing Line ✓** B. Hover Line C. GPS Line D. Orbit Line

Why: The racing line is the fastest path through a corner. Hover, GPS and orbit lines are not racing terms. Example: taking a wide entry and a late apex keeps speed up through the corner.

PILOT-008

What is the primary goal of cinematic FPV flying?

- A. Maximum Speed **B. Smooth Storytelling Through Motion ✓** C. Maximum Altitude D. Fastest Lap Time

Why: The goal of cinematic FPV flying is smooth storytelling through motion — shots that look purposeful and flowing. Maximum speed, altitude or lap time are racing goals. Example: a long, flowing reveal around a building.

PILOT-009

Fast pilots focus on:

- A. Exit Point ✓** B. Drone C. Props D. OSD

Why: Fast pilots focus on the exit point of a turn, not the drone, props or OSD, so they set up speed for what follows. Example: looking at where the corner leads rather than at the drone.

PILOT-010

The key to smooth dives is:

- A. Controlled throttle management ✓** B. Full throttle C. GPS lock D. Higher camera angle

Why: Smooth dives need controlled throttle management so the drone does not speed up too much or bounce. Full throttle, GPS lock and camera angle do not give a smooth dive. Example: ease the throttle as you drop down a cliff face.

PILOT-011

During an orbit shot the subject should remain:

- A. Center Frame ✓** B. Behind Drone C. Above Drone D. Below Drone

Why: During an orbit shot the subject should stay in the centre of the frame while the drone circles it. Behind, above or below is wrong. Example: a circle around a car with the car held in the middle of the picture.

PILOT-012

The fastest race drone is useless without:

- A. Pilot Skill ✓** B. Battery C. GPS D. OSD

Why: The fastest race drone is useless without pilot skill, since the pilot decides how the machine is flown. Battery, GPS and OSD do not replace skill. Example: a top-end quad flown badly loses to a modest quad flown well.

PILOT-013

The best way to fly through a gap is:

- A. Focus beyond the gap ✓** B. Look at obstacle C. Close eyes D. Full throttle

Why: Focus beyond the gap, at where you want to go, not at the obstacle. Looking at the obstacle makes you fly into it. Example: look through the gate toward the next target, and the drone follows.

PILOT-014

Smooth drifts require:

- A. Controlled roll and yaw ✓** B. Full throttle C. GPS D. Brake mode

Why: Smooth drifts need controlled roll and yaw working together. Full throttle, GPS or brake mode will not make a clean drift. Example: yaw the nose into the turn while holding the bank.

PILOT-015

The purpose of a Figure-8 drill is:

- A. Learn coordinated turns ✓** B. Learn GPS C. Learn batteries D. Learn charging

Why: A figure-8 drill teaches coordinated turns in both directions, linking roll, pitch and yaw. It is not about GPS, batteries or charging. Example: practising smooth figure-8s builds control.

PILOT-016

What is a coordinated turn?

- A. Roll + Pitch + Small Yaw ✓** B. Throttle Only C. Pitch Only D. Roll Only

Why: A coordinated turn uses roll, pitch and a small amount of yaw together so the drone flows through the arc. Throttle-, pitch- or roll-only turns skid. Example: bank, pull a little pitch and add a touch of yaw.

PILOT-017

Target tracking requires:

- A. Prediction ✓** B. Aggression C. Random Inputs D. High VTX

Why: Target tracking requires prediction — anticipating where the subject will be next. Aggression, random inputs or a high VTX power do not help. Example: aim ahead of a moving car rather than at where it is now.

PILOT-018

What makes drone motion appear professional?

- A. Predictable movement ✓** B. Random movement C. Fast movement D. Aggressive movement

Why: Professional-looking drone motion is predictable: smooth, planned and consistent. Random, fast or aggressive movement looks amateur. Example: viewers enjoy a steady, planned camera move.

PILOT-019

Which movement causes sideways flight?

- A. Roll ✓ B. Pitch C. Throttle D. GPS

Why: Roll tilts the drone sideways, producing sideways flight. Pitch gives forward or backward, and throttle controls height. Example: push the right stick right to slide sideways.

PILOT-020

Advanced pilots think:

- A. Ahead of the Aircraft ✓ B. At the Aircraft C. Behind the Aircraft D. Randomly

Why: Advanced pilots think ahead of the aircraft, planning the next move. Thinking at or behind the aircraft is reactive. Example: planning the next three gates while flying the current one.

PILOT-021

The best pilots use:

- A. Minimal Inputs ✓ B. Maximum Inputs C. Random Inputs D. Aggressive Inputs

Why: The best pilots use minimal inputs, small precise stick movements. Maximum, random or aggressive inputs make the flight rough. Example: tiny corrections keep the quad steady.

PILOT-022

Why do beginners wobble during turns?

- A. Overcorrection ✓ B. Strong Motors C. GPS D. Antennas

Why: Beginners wobble in turns due to overcorrection — big stick moves that overshoot, then correct back. Strong motors, GPS or antennas are not the cause. Example: make smaller, calmer stick inputs.

PILOT-023

The best tracking pilots think:

- A. Several Seconds Ahead ✓ B. One Frame Ahead C. Randomly D. Only Present Moment

Why: The best tracking pilots think several seconds ahead, anticipating the subject's path. One frame ahead or only the present is too late. Example: anticipate a rider's line through a corner.

PILOT-024

Good framing means:

- A. Subject Positioned Intentionally ✓ B. Random Subject Placement C. Maximum Zoom D. Fast Motion

Why: Good framing means the subject is positioned intentionally in the shot. Random placement or maximum zoom is not good framing. Example: placing the subject on a third of the frame on purpose.

PILOT-025

Tracking skill improves through:

- A. Repetition ✓ B. Hardware C. Batteries D. Antennas

Why: Tracking skill improves through repetition — practising again and again. Hardware, batteries or antennas do not build skill. Example: tracking a moving car many times builds feel for it.

PILOT-026

What creates jerky video?

- A. Abrupt stick inputs ✓** B. GPS C. OSD D. Battery

Why: Jerky video comes from abrupt stick inputs: a sudden stick movement changes the drone's attitude instantly, and the camera shows every jolt. Hardware or GPS do not cause it. Example: snapping the stick to start a turn makes the footage lurch, while easing the stick in and out, or adding expo, gives smooth motion.

PILOT-027

What improves gate accuracy?

- A. Repetition ✓** B. Bigger Motors C. Bigger Battery D. Higher VTX

Why: Gate accuracy improves through repetition, practising the same line. Bigger motors, bigger batteries or higher VTX power do not. Example: repeating a gate sequence until it is automatic.

PILOT-028

Why do professional pilots look ahead?

- A. Predict future path ✓** B. Avoid flying C. Reduce speed D. Save battery

Why: Professional pilots look ahead to predict their future path and plan the line. It is not to avoid flying, reduce speed or save battery. Example: looking at the next corner while passing the current one.

PILOT-029

The biggest mistake in tracking is:

- A. Chasing Current Position ✓** B. Predicting Movement C. Smooth Flying D. Looking Ahead

Why: The biggest mistake in tracking is chasing the current position, which always lags behind the subject. Predicting, smooth flying and looking ahead are good habits. Example: aim where the subject will be, not where it is.

PILOT-030

Altitude holding skill depends on:

- A. Throttle Discipline ✓** B. Battery Size C. GPS D. Motors

Why: Altitude holding depends on throttle discipline: steady, small throttle control. Battery size, GPS or motors do not matter as much. Example: hold the throttle steady to stay at one height.

PILOT-031

Professional pilots recover from mistakes by:

- A. Small corrections ✓** B. Large corrections C. Panic inputs D. Disarming

Why: Professionals recover from mistakes with small corrections, staying calm. Large, panic inputs or disarming make things worse. Example: a small stick nudge brings the quad back level.

PILOT-032**Muscle memory develops from:**

- A. Repetition ✓** B. Watching Videos C. Buying Equipment D. Reading Manuals Only

Why: Muscle memory develops from repetition — doing the movement many times. Watching videos, buying gear or reading manuals alone do not build it. Example: hours of simulator practice make stick control automatic.

PILOT-033**Smooth footage depends mostly on:**

- A. Pilot Skill ✓** B. Battery Brand C. Antenna Type D. Frame Color

Why: Smooth footage depends mostly on pilot skill. Battery brand, antenna type or frame colour matter little. Example: a skilled pilot with modest gear still produces smooth video.

PILOT-034**During takeoff, throttle should be:**

- A. Smoothly increased ✓** B. Instantly maxed C. Constantly pulsed D. Disabled

Why: During takeoff the throttle should be smoothly increased so the drone lifts off controlled. Instantly maxed, pulsed or disabled throttle is unsafe. Example: raise the stick gradually until the quad is light on its feet.

PILOT-035**What stick rotates the drone left or right?**

- A. Roll B. Pitch **C. Yaw ✓** D. Throttle

Why: Yaw rotates the drone left or right about its vertical axis (usually the left stick horizontally). Roll tilts sideways, pitch tilts forward/back and throttle changes height. Example: push the left stick left to turn the nose left.

PILOT-036**A late apex usually provides:**

- A. Better Exit ✓** B. Worse Exit C. Slower Turn D. More Wobble

Why: A late apex (turning in later) usually gives a better exit, so you can accelerate earlier out of the turn. It does not give a worse exit or more wobble. Example: turn in later and power out of the corner straighter.

PILOT-037**The best practice for learning FPV is:**

- A. Acro mode immediately **B. Simulator first ✓** C. Full speed flights D. Racing first day

Why: The best practice for learning FPV is a simulator first, where crashes cost nothing. Acro immediately, full-speed flights or racing on day one lead to crashes. Example: spend hours in the simulator before flying a real quad.



PILOT-038

Which movement causes forward flight?

- A. Roll **B. Pitch Forward ✓** C. Yaw D. Throttle Only

Why: Pitching the nose forward causes forward flight. Roll moves sideways, yaw turns, and throttle alone changes height. Example: push the right stick forward to fly forward.

PILOT-039

The fastest pilots are:

- A. Efficient ✓** B. Aggressive C. Reckless D. Lucky

Why: The fastest pilots are efficient: smooth lines, minimal inputs, no wasted motion. Aggressive, reckless or lucky pilots are not consistently fast. Example: smooth lines beat sharp, jerky ones.

PILOT-040

Race performance improves most from:

- A. Practice ✓** B. Hardware C. Batteries D. VTX

Why: Race performance improves most from practice, because speed comes from muscle memory, line choice and consistency, which only come from repetition. Hardware, batteries and VTX help only slightly. Example: a racer who flies the same track for 30 minutes every day will cut more time than one who buys a new set of motors.

PILOT-041

Good tracking footage appears:

- A. Effortless ✓** B. Aggressive C. Jerky D. Random

Why: Good tracking footage looks effortless: the subject stays framed and the motion is smooth and relaxed. Aggressive, jerky or random footage looks amateur. Example: a steady follow of a cyclist where the camera appears to glide.

PILOT-042

What is an Orbit Shot?

- A. Flying Circles Around Subject ✓** B. Hovering C. Landing D. Racing

Why: An orbit shot is flying circles around a subject, keeping the camera pointed at it. Hovering, landing or racing are different. Example: circling a statue while keeping it centred.

PILOT-043

Which skill is most important for target tracking?

- A. Relative Position Awareness ✓** B. Maximum Speed C. Bigger Battery D. High KV Motors

Why: The most important skill for target tracking is relative position awareness: knowing where you are compared with the subject. Maximum speed, bigger batteries and high-KV motors do not help. Example: judging distance and angle to a moving car.

PILOT-044

Professional pilots maintain altitude by:

- A. Continuous micro-adjustments ✓** B. Full throttle bursts C. GPS only D. Ignoring throttle

Why: Professionals maintain altitude with continuous micro-adjustments of the throttle. Full throttle bursts, GPS only or ignoring the throttle cause bobbing. Example: tiny throttle nudges hold a steady height.

PILOT-045

Closing Velocity means:

- A. Rate Distance Shrinks ✓** B. Motor RPM C. Battery Voltage D. GPS Speed

Why: Closing velocity is the rate at which the distance between you and the target shrinks. It is not motor RPM, battery voltage or GPS speed. Example: if the gap to a car closes by 5 metres every second, closing velocity is 5 m/s.

PILOT-046

The best way to learn precision flying is:

- A. Repeat drills daily ✓** B. Buy larger motors C. Use stronger VTX D. Fly randomly

Why: The best way to learn precision flying is to repeat drills daily. Bigger motors, stronger VTX or flying randomly do not build precision. Example: daily gate and hover drills sharpen control.

PILOT-047

Good FPV pilots constantly monitor:

- A. Position, Speed, Altitude ✓** B. Motors Only C. Battery Only D. Camera Only

Why: Good FPV pilots constantly monitor position, speed and altitude. Watching motors only, battery only or camera only gives an incomplete picture. Example: glance at the OSD while watching the flight line.

PILOT-048

Moving targets require:

- A. Anticipation ✓** B. Guessing C. Luck D. Hovering

Why: Moving targets require anticipation — predicting where they will go. Guessing, luck or hovering do not work. Example: anticipate a car's line through a corner and aim ahead.

PILOT-049

When following a moving car, pilots should:

- A. Predict movement ✓** B. Chase aggressively C. Ignore speed D. Fly randomly

Why: When following a moving car, predict its movement and plan ahead. Chasing aggressively, ignoring speed or flying randomly leads to poor footage. Example: anticipate braking and corners.

PILOT-050

Professional pilots avoid:

- A. Tunnel Vision ✓** B. Situational Awareness C. Planning D. Prediction

Why: Professional pilots avoid tunnel vision — focusing on one thing and losing awareness of the rest. Situational awareness, planning and prediction are good habits. Example: keep an eye on obstacles around you while tracking.

PILOT-051

The best method to eliminate wobbling is:

- A. Smaller stick corrections ✓** B. Bigger corrections C. Full throttle D. More yaw

Why: The best way to eliminate wobbling is smaller stick corrections. Bigger corrections, full throttle or more yaw make it worse. Example: calm, tiny stick moves stop the oscillation.

PILOT-052

What is a Reveal Shot?

- A. Gradually Showing Subject ✓** B. Fast Dive C. Crash Recovery D. Hover

Why: A reveal shot gradually shows the subject, for example by flying past an obstacle or rising to uncover a view. A fast dive, crash recovery or hover is not a reveal. Example: rising over a ridge to reveal a valley.

PILOT-053

For cinematic footage throttle should be:

- A. Smooth ✓** B. Aggressive C. Random D. Maximum

Why: For cinematic footage the throttle should be smooth. Aggressive, random or maximum throttle makes the video jerky. Example: gentle throttle changes keep the shot even.

PILOT-054

Pure Pursuit means:

- A. Point Directly At Target ✓** B. Fly Away C. Hover D. Orbit

Why: Pure pursuit means pointing directly at the target and following it. Flying away, hovering or orbiting is different. Example: always aiming the nose at a moving car.

PILOT-055

A perfect lap is:

- A. Smooth and Consistent ✓** B. Aggressive C. Random D. Lucky

Why: A perfect lap is smooth and consistent: smooth lines keep speed up, and consistency means you can repeat it. An aggressive, random or lucky lap cannot be repeated. Example: a lap with no corrections and the same speed through every gate beats a faster but scrappy lap that sometimes ends in a crash.



PILOT-056

Overshooting a gate usually means:

- A. Poor Corner Setup ✓** B. GPS Failure C. ESC Failure D. Camera Failure

Why: Overshooting a gate usually means poor corner setup — wrong entry speed or line. GPS, ESC or camera failures are not the typical cause. Example: set up the turn earlier to line up the next gate.

PILOT-057

The biggest beginner mistake is:

- A. Looking Too Close ✓** B. Looking Ahead C. Smooth Flying D. Planning Turns

Why: The biggest beginner mistake is looking too close (at the drone or the obstacle) instead of ahead. Looking ahead, smooth flying and planning turns are good. Example: look further down the course.

PILOT-058

Why should stick movements be small?

- A. Better control ✓** B. More crashes C. More speed D. More battery

Why: Stick movements should be small because the drone responds in proportion to how far the stick moves. Small inputs give small, precise changes, while big inputs cause big movements that overshoot and need correcting. Example: a gentle nudge banks the quad a few degrees to line up with a gap, while a full stick throw sends it sliding past.

PILOT-059

The ultimate FPV skill is:

- A. Precise Intentional Control ✓** B. Maximum Speed C. Highest Altitude D. Biggest Battery

Why: The ultimate FPV skill is precise, intentional control, doing exactly what you meant. Maximum speed, height or a big battery are not skills. Example: placing the quad exactly where you planned.

PILOT-060

What is the safest first maneuver?

- A. Power Loop **B. Hover ✓** C. Dive D. Split-S

Why: Hover is the safest first manoeuvre, as it teaches basic control with little risk. Power loops, dives and split-S are advanced. Example: practise hover in the simulator first.

PILOT-061

What is the purpose of a hover exercise?

- A. Learn throttle control ✓** B. Learn GPS C. Learn batteries D. Learn ESC

Why: A hover exercise teaches throttle control, holding a steady height. It is not for learning GPS, batteries or ESCs. Example: hovering at one height trains throttle feel.

PILOT-062

The biggest reason for wobble after a turn is:

- A. Excessive correction input ✓** B. Battery Voltage C. GPS D. OSD

Why: The biggest reason for wobble after a turn is excessive correction input — overcorrecting stick movements. Battery voltage, GPS and OSD are not the cause. Example: smaller exit corrections settle the quad.

PILOT-063

When entering a turn at high speed:

- A. Plan exit before entry ✓** B. React at last second C. Full yaw D. Full roll

Why: Entering a turn at high speed, plan the exit before the entry so you set up the right line and speed. Reacting at the last second, full yaw or full roll leads to errors. Example: decide how you leave the corner before turning in.

PILOT-064

A professional chase shot requires:

- A. Constant framing and distance ✓** B. Maximum speed C. Random movement D. Aggressive yaw

Why: A professional chase shot requires constant framing and distance to the subject. Maximum speed, random movement or aggressive yaw spoil the shot. Example: stay a steady distance behind the subject.

PILOT-065

What stick controls left-right tilt?

- A. Pitch **B. Roll ✓** C. Throttle D. GPS

Why: Roll controls left-right tilt. In Mode 2 it is the right stick moved sideways. Pitch (right stick forward/back) tilts forward and back, and throttle controls height. Example: pushing the right stick to the right drops the right side of the quad and makes it slide to the right.

PILOT-066

Professional racers prioritize:

- A. Exit Speed ✓** B. Entry Speed C. Top Speed D. Battery Life

Why: Professional racers prioritise exit speed, because speed leaving a corner carries down the next straight. Entry speed or top speed alone do not win laps. Example: a slower entry with a faster exit is quicker overall.

PILOT-067

What is the easiest landing technique?

- A. Full throttle drop **B. Slow descent ✓** C. Disarm mid-air D. Dive landing

Why: The easiest landing technique is a slow descent: reduce throttle gradually and touch down gently. Full throttle drops, disarming mid-air or dive landings are dangerous. Example: ease down smoothly.

PILOT-068

Lead Room means:

- A. Space Ahead of Subject ✓** B. Space Behind Subject C. Empty Sky D. GPS Data

Why: Lead room is space ahead of the subject in the frame, in the direction it is moving. It is not space behind, empty sky or GPS data. Example: leave space in front of a moving car.

PILOT-069

Lead Pursuit means:

- A. Aim Ahead of Target ✓** B. Aim Behind Target C. Aim Below Target D. Aim Randomly

Why: Lead pursuit means aiming ahead of the target, at the point where it is going to be, so that you meet it instead of trailing behind. Example: to track a car crossing in front of you, point the nose in front of it and fly toward that spot; chasing its current position always leaves you behind.

PILOT-070

Smooth flying requires:

- A. Muscle Memory ✓** B. Bigger Battery C. GPS D. Faster ESC

Why: Smooth flying requires muscle memory from practice. Bigger batteries, GPS or faster ESCs do not provide it. Example: hours of practice make smooth stick control automatic.

PILOT-071

The purpose of race lines is:

- A. Minimize Lap Time ✓** B. Increase Battery Life C. Improve GPS D. Increase Altitude

Why: Race lines exist to minimise lap time: the line is the path that lets you carry the most speed through the course. Example: through a fast right-hand corner you enter wide, clip the inside at the apex and exit wide; that path needs less braking and is quicker than cutting tightly to the inside.

PILOT-072

Interceptor pilots focus on:

- A. Intercept Point ✓** B. Current Position C. Battery D. OSD

Why: Interceptor pilots focus on the intercept point — where the paths will meet. Current position, battery and OSD are not the focus. Example: aim at the predicted meeting point.

PILOT-073

Professional hovering means maintaining position within:

- A. Small Defined Area ✓** B. Entire Field C. Random Position D. GPS Range

Why: Professional hovering means staying within a small defined area. The entire field, random positions or GPS range are too loose. Example: hold position within a metre.

PILOT-074

Parallax Effect occurs when:

- A. Foreground Moves Faster Than Background ✓** B. Everything Moves Same Speed C. Drone Stops D. GPS Fails

Why: The parallax effect occurs when the foreground moves faster than the background as the camera moves. It is not everything moving at the same speed. Example: close trees blur past while distant mountains barely move.

PILOT-075

To hover at the same height, the pilot should:

- A. Hold constant throttle ✓** B. Hold maximum throttle C. Hold minimum throttle D. Disable motors

Why: To hover at the same height, hold a constant throttle (with tiny adjustments). Maximum, minimum or disabled throttle changes height. Example: keep the throttle steady at the hover point.

PILOT-076

The most common mistake in cinematic FPV is:

- A. Overcorrection ✓** B. Too Smooth Flying C. Low VTX Power D. Low Battery

Why: The most common mistake in cinematic FPV is overcorrection, which makes the video wobble. Flying too smoothly or low VTX power are not mistakes. Example: smaller corrections keep shots steady.

PILOT-077

Professional tracking requires:

- A. Constant Relative Position ✓** B. Constant Battery C. Constant Altitude D. Constant Yaw

Why: Professional tracking requires keeping a constant relative position to the subject. Constant battery, altitude or yaw are not required. Example: maintain the same distance and angle.

PILOT-078

The best racers make:

- A. Minimal Corrections ✓** B. Large Corrections C. Random Corrections D. Panic Inputs

Why: The best racers make minimal corrections because they fly a planned line, so little fixing is needed. Every correction costs speed and can start a wobble. Example: setting up a turn early lets you make one smooth input, while a late entry means stabbing the sticks several times to recover.

PILOT-079

The biggest enemy of cinematic footage is:

- A. Jerky Inputs ✓** B. Low Wind C. High Battery D. GPS

Why: Jerky inputs are the biggest enemy of cinematic footage, because every sudden stick movement shows up on screen. Hardware cannot fully hide bad technique. Example: ramp the sticks gently, use expo for soft response near the centre, and the same quad that gave shaky video produces smooth footage.

PILOT-080

What creates professional-looking footage?

- A. Smooth Acceleration B. Smooth Deceleration C. Smooth Turns **D. All of the Above ✓**

Why: Smooth acceleration, smooth deceleration and smooth turns all contribute to professional-looking footage, so the answer is all of the above. Example: ease into a move, cruise, then ease out, like a camera on a rail; any one abrupt part of the move breaks the flow.

PILOT-081

What causes over-shooting a target?

- A. Poor anticipation ✓** B. GPS C. Camera Settings D. Battery Type

Why: Overshooting a target is caused by poor anticipation: not judging how far the drone will travel while you slow down. GPS, camera settings and battery type do not cause it. Example: begin slowing earlier before reaching a gate.

PILOT-082

Professional cinematic pilots often use:

- A. Expo and Rates Tuning ✓** B. Maximum Rates C. Racing Rates D. Default Settings Only

Why: Cinematic pilots often tune expo and rates to get soft, smooth stick response around centre. Maximum, racing or default-only rates feel twitchy. Example: add expo so small stick movements make gentle changes.

PILOT-083

Which skill improves race line efficiency?

- A. Corner Entry Planning ✓** B. More Power C. Bigger Props D. Louder Beeper

Why: Corner entry planning improves race line efficiency, because the entry decides the line and exit. More power, bigger props or a louder beeper do not. Example: plan the line before turning in.

PILOT-084

The apex is:

- A. Inside Point of Turn ✓** B. Landing Point C. Start Gate D. Finish Gate

Why: The apex is the inside point of a turn, the spot closest to the inside of the corner. It is not the landing point or the start or finish gate. Example: aim to clip the apex and then accelerate out.

PILOT-085

Good racers fly:

- A. Predictable Lines ✓** B. Random Lines C. Wide Lines Only D. Tight Lines Only

Why: Good racers fly predictable lines: the same path every lap. That lets them time their inputs, build muscle memory and push speed safely. Random lines force constant correction. Example: hitting the same apex on every lap lets you commit to the speed because you know exactly what will happen.

PILOT-086

Wind compensation requires:

- A. Anticipation ✓** B. Guessing C. Aggression D. Hovering

Why: Wind compensation requires anticipation: expect the gust and correct early. Guessing, aggression or hovering do not help. Example: lean into a crosswind before it pushes you off line.

PILOT-087

The fastest racing pilots focus on:

- A. Smoothness over aggression ✓** B. Maximum aggression C. Random corrections D. Bigger batteries

Why: The fastest racing pilots focus on smoothness over aggression. Maximum aggression, random corrections or bigger batteries are slower. Example: smooth lines carry more speed.

PILOT-088

The smoothest FPV flights are achieved through:

- A. Precision and anticipation ✓** B. Raw speed C. Bigger motors D. Higher VTX

Why: The smoothest flights come from precision (small, accurate stick inputs) and anticipation (planning the move before you need it). Raw speed, bigger motors or higher VTX power do not make flying smooth. Example: starting a turn early with a tiny input instead of late with a big one.

PILOT-089

What controls orbit radius?

- A. Roll + Yaw Balance ✓** B. Throttle Only C. Pitch Only D. GPS

Why: Orbit radius is controlled by the balance of roll and yaw: how much you bank versus how fast you rotate. Throttle-only, pitch-only or GPS do not set it. Example: more yaw with the same roll tightens the circle.

PILOT-090

Professional flying prioritizes:

- A. Control ✓** B. Speed C. Power D. Hardware

Why: Professional flying prioritises control, because speed without control ends in crashes. Pros add speed only as far as they can still control it. Example: a controlled pilot flying at 80% pace finishes the race, while a reckless pilot at 100% crashes on lap two.

PILOT-091

The largest time loss usually occurs from:

- A. Poor Corners ✓** B. Straight Sections C. Takeoff D. Landing

Why: The largest time loss usually occurs from poor corners, where speed is lost and the line is wrong. Straights, takeoff and landing lose little. Example: losing 0.3 seconds in each corner adds up fast.

PILOT-092

What causes sudden altitude jumps?

- A. Aggressive throttle inputs ✓** B. GPS C. Camera angle D. VTX power

Why: Sudden altitude jumps come from aggressive throttle inputs: throttle sets total thrust, so a quick stab on the stick changes thrust and height immediately. GPS, camera angle or VTX power do not. Example: punching the throttle at the bottom of a dive makes the quad leap upward; easing the stick brings it out smoothly.

PILOT-093

Fast racers fly:

- A. Smooth ✓** B. Aggressive C. Random D. Unpredictable

Why: Fast racers fly smooth, because smooth lines carry speed through corners and avoid wasted corrections. "Smooth is fast". Example: a smooth arc through a gate keeps the speed up, while an aggressive snap turn scrubs speed and risks a crash.

PILOT-094

A master FPV pilot can:

- A. Place the Drone Exactly Where Intended Every Time ✓** B. Fly Fast Only C. Hover Only D. Race Only

Why: A master FPV pilot can place the drone exactly where intended every time. Flying fast only, hovering only or racing only are narrower. Example: threading the same tiny gap repeatedly.

PILOT-095

What is the primary purpose of racing lines?

- A. Minimize time through course ✓** B. Increase battery weight C. Improve GPS D. Reduce VTX power

Why: The primary purpose of racing lines is to minimise time through the course. They do not increase battery weight, improve GPS or reduce VTX power. Example: choose the shortest, fastest path.

PILOT-096

What separates a professional FPV pilot from a beginner?

- A. Ability to consistently place the drone exactly where intended ✓** B. Bigger batteries C. Faster motors D. Stronger antennas

Why: A professional is separated from a beginner by the ability to consistently place the drone exactly where intended. Bigger batteries, faster motors or stronger antennas do not. Example: pros hit the line they choose.

PILOT-097

The best orbit speed is:

- A. Constant ✓** B. Random C. Maximum D. Minimum

Why: The best orbit speed is constant. A steady speed and a steady radius give an even circle and keep the subject in the same place in the frame. Changing speed makes the subject drift and the shot look uneven. Example: orbiting a monument at a steady 5 m/s gives a clean, professional circle.

PILOT-098

Professional pilots use yaw:

- A. Sparingly ✓** B. Constantly C. Aggressively D. Randomly

Why: Professional pilots use yaw sparingly. Yaw rotates the nose and camera quickly, so heavy yaw makes the video jerky. Most turning is done with roll, and a small amount of yaw refines the heading. Example: a touch of yaw keeps the subject centred during an orbit instead of sweeping the camera around.

PILOT-099

What improves altitude consistency?

- A. Throttle Discipline ✓** B. Bigger Battery C. GPS D. Antenna

Why: Altitude consistency comes from throttle discipline: holding a steady throttle and making only tiny adjustments. Battery size, GPS or antennas do not fix bobbing up and down. Example: practise hovering at 2 metres for a whole battery without climbing or sinking.

PILOT-100

During descent, pilots should:

- A. Avoid own prop wash ✓** B. Disable throttle C. Max throttle D. Ignore air flow

Why: During descent, pilots should avoid their own prop wash, which makes the drone unstable. Disabling or maxing throttle, or ignoring airflow, is dangerous. Example: descend with some forward motion.

PILOT-101

A professional chase shot maintains:

- A. Constant Distance B. Constant Altitude C. Consistent Framing **D. All of the Above ✓**

Why: A professional chase shot keeps a constant distance, a constant altitude and consistent framing, so all three together. Example: when following a mountain biker you stay about 10 metres behind and 3 metres up, with the rider in the same spot in the frame, so the shot looks calm and professional.

PILOT-102

What stick primarily controls altitude?

- A. Roll B. Pitch **C. Throttle ✓** D. Yaw

Why: Throttle controls altitude (in Mode 2 it is the left stick moved up and down). It sets the total thrust: more thrust than the drone's weight climbs, less descends, and equal holds a hover. Example: raise the stick to climb and lower it gently to descend; roll, pitch and yaw only tilt and rotate the drone.

PILOT-103

Professional racers anticipate:

- A. Next Gate ✓** B. Current Gate C. OSD D. Battery

Why: Professional racers anticipate the next gate, so they can set up the line. The current gate, OSD or battery are not the focus. Example: look ahead and plan the next gate.

PILOT-104

A professional pilot's most important asset is:

- A. Situational Awareness ✓** B. Battery C. VTX D. Camera

Why: A professional pilot's most important asset is situational awareness. Battery, VTX and camera are equipment. Example: knowing where the drone, obstacles and people are.

PILOT-105

Professional landings are:

- A. Controlled ✓** B. Aggressive C. Fast D. Random

Why: Professional landings are controlled: a slow, gentle descent with the drone level, ending in a soft touchdown. Aggressive, fast or random landings risk breaking props, arms or the camera. Example: reduce the throttle gradually over the last metre, touch down, then disarm on the ground, never in the air.

PILOT-106

For cinematic shots, turns should be:

- A. Wide and smooth ✓** B. Sharp C. Aggressive D. Instant

Why: For cinematic shots, turns should be wide and smooth. Sharp or instant turns make the camera whip and the horizon tilt, which looks jerky. Wide arcs give natural, flowing motion. Example: sweeping in a long curve around a building looks cinematic, while a tight snap turn looks like a racing clip.

PILOT-107

What skill prevents crashes most?

- A. Orientation Awareness ✓** B. Fast Motors C. High VTX D. GPS

Why: Orientation awareness prevents the most crashes, because you know which way the drone is facing. Fast motors, high VTX and GPS do not. Example: know which way is forward.

PILOT-108

Professional pilots look:

- A. Ahead of Current Position ✓** B. At Drone C. At Ground D. At OSD Only

Why: Professional pilots look ahead of their current position so they have time to plan the line and spot obstacles. Looking only at the drone, the ground or the OSD makes you react too late. Example: while flying through one gate, scan the next two gates so you already know the line.

PILOT-109

A perfect orbit shot requires:

- A. Constant radius around target ✓** B. Random radius C. Random speed D. Full yaw

Why: A perfect orbit needs a constant radius around the target. If the radius changes, the subject grows and shrinks in the frame and the shot looks wobbly. Example: pick a 10-metre circle around a tree and hold that same distance all the way round, using a steady speed.

PILOT-110

A Reveal Shot creates:

- A. Surprise B. Storytelling C. Scale **D. All of the Above ✓**

Why: A reveal shot creates surprise (the subject is hidden then shown), storytelling (it builds anticipation) and a sense of scale (it shows how big the subject is), so all of the above. Example: rising slowly from behind a hill to reveal a castle in the valley.

PILOT-111

For cinematic flying, stick movement should be:

- A. Smooth ✓** B. Aggressive C. Random D. Maximum

Why: For cinematic flying, stick movement should be smooth. Gradual stick movement produces gradual changes in the drone, which show up as steady video. Example: move the stick as if squeezing, not flicking, and use expo to soften the response around the centre.

PILOT-112

Maintaining target center frame requires:

- A. Continuous Small Corrections ✓** B. Large Corrections C. Random Inputs D. Full Roll

Why: Keeping the target centred requires continuous small corrections. Large corrections, random inputs or full roll make it drift. Example: tiny nudges keep the subject in the middle.

PILOT-113

Which pursuit reaches moving targets fastest?

- A. Lead Pursuit ✓** B. Lag Pursuit C. Hover Pursuit D. Random Pursuit

Why: Lead pursuit reaches moving targets fastest, because you aim ahead of where they will be. Lag, hover or random pursuit are slower. Example: aim ahead of a moving car.

PILOT-114

What is throttle modulation?

- A. Fine throttle adjustments ✓** B. Full throttle C. Motor replacement D. ESC tuning

Why: Throttle modulation is fine throttle adjustment — small changes to control height and speed. It is not full throttle, motor replacement or ESC tuning. Example: gentle throttle changes in a descent.

PILOT-115

What causes prop wash oscillations?

- A. Turbulent air ✓** B. GPS C. VTX D. Receiver

Why: Prop wash oscillations are caused by turbulent air. When you descend quickly or slow down sharply, the quad flies through its own disturbed air, so the props get inconsistent airflow and thrust, which makes it wobble. Example: the shaking at the end of a dive; easing the descent and good filter tuning reduce it.

PILOT-116

The secret of cinematic flying is:

- A. Predictability ✓** B. Randomness C. Aggression D. Speed

Why: The secret of cinematic flying is predictability: planned, repeatable, smooth motion that viewers (and clients) can follow. Randomness, aggression and speed work against it. Example: a planned route flown the same way twice gives a shot you can match and edit together.

PILOT-117

Which maneuver teaches smooth transitions?

- A. Figure-8 ✓** B. Crash Recovery C. Dive D. Split-S

Why: The figure-8 teaches smooth transitions, because you continually switch from a left turn to a right turn, changing the bank angle smoothly while holding speed and height. Crash recovery, dives and split-S are different skills. Example: practise a figure-8 around two cones until the changeover is effortless.

PILOT-118

When learning, altitude should be:

- A. High enough to recover ✓** B. 1 cm above ground C. Maximum possible D. Touching trees

Why: When learning, altitude should be high enough to recover from mistakes. 1 cm above ground, maximum possible or touching trees is unsafe or useless. Example: a few metres of height gives time to correct.

PILOT-119

Professional recovery means:

- A. Correcting Early ✓** B. Correcting Late C. Panic Inputs D. Disarming

Why: Professional recovery means correcting early, before an error grows. Correcting late, panic inputs or disarming make things worse. Example: catch a drift early with a small input.

PILOT-120

Fast targets require:

- A. Lead Angle ✓** B. Hovering C. Orbiting D. Landing

Why: Fast targets require a lead angle: you aim ahead of the target by an amount that grows with its speed, so you meet it. Hovering, orbiting or landing do not track it. Example: for a fast-moving car you point the nose well ahead of where it is now, not at the car itself.

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